

nRF9160 DK Hardware

v1.1.0

User Guide

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Revision history

Date	Description
May 2024	Added: • Environmental and safety notices on page vi Updated: • Debug out for programming external boards on page 29 • Programming an external board on page 29 • Programming a board with custom connections on page 31 Editorial changes
April 2022	• Updated to match nRF9160 DK v1.1.0 • Editorial changes
October 2021	• Updated Board control on page 20 • Editorial changes
November 2020	Updated to match nRF9160 DK v0.15.0 and nRF9160 DK v1.0.0 • Updated: • Kit content on page 8 • Figure 7: nRF9160 DK block diagram on page 13 • Firmware development mode on page 9 • Performance measurement mode on page 10 • Hardware description on page 12 • GNSS on page 17 • Board control on page 20 • New: • I/O expander on page 25 • External memory on page 27 • Switches for I/O expander on page 37 • Switches for external memory on page 36
July 2020	Updated document title
April 2020	Editorial changes
March 2020	• Updated Recommended reading on page 50 • Updated Debug output • Added Connectors for programming external boards
October 2019	Updated to match nRF9160 DK v0.9.0 • Updated: • Kit content on page 8 • Recommended reading on page 50 • Firmware development mode on page 9

Date	Description
	• Performance measurement mode on page 10 • Block diagram on page 12 • Hardware drawings on page 12 • Power supply on page 13 • Antenna interfaces on page 17 • GNSS on page 17 • Board control on page 20 • SIM and eSIM on page 37 • Additional interfaces on page 38 • Solder bridge configuration on page 39 • Prepare the DK for current measurements on page 41 • New: • Interface MCU disconnect switches on page 33 • Switches for UART interface on page 33 • Switches for nRF52840 interface on page 35 • Removed: • Getting started
July 2019	Updated: • nRF9160 power supply on page 16
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December 2018	Updated to match nRF9160 DK v0.8.2
November 2018	Preview DK changed into DK
October 2018	First release

Previous versions

PDF files for the previous versions are available here:

- [nRF9160 DK Hardware User Guide v1.0.0](#)
- [nRF9160 DK Hardware User Guide v0.9.3](#)
- [nRF9160 DK User Guide v0.9](#)
- [nRF9160 DK User Guide v0.7.2](#)
- [nRF9160 DK User Guide v0.7.1](#)
- [nRF9160 DK User Guide v0.7](#)

Environmental and safety notices

Environmental and safety notices for the DK and power supply requirements.

Note: The nRF9160 DK must be powered by a PS1 class (IEC 62368-1) power supply with maximum power less than 15 W.

Skilled persons

The nRF9160 DK is intended for use only by skilled persons.

A skilled person is someone with relevant education or experience that enables them to identify potential hazards and takes appropriate action to reduce the risk of injury to themselves and others.



Electrostatic discharge

The nRF9160 DK is susceptible to *Electrostatic Discharge (ESD)*.

To avoid damage to your device, it should be used in an electrostatic free environment, such as a laboratory.



Environmental Protection

Waste electrical products should not be disposed of with household waste.

Please recycle where facilities exist. Check with your local authority or retailer for recycling advice.

1 Introduction

The nRF9160 DK is a hardware development platform used to design and develop application firmware on the nRF9160 *System in Package (SiP)*. The nRF9160 SiP supports *Long-Term Evolution (LTE) Cat-M1* and *Cat-NB1*.

The *Development Kit (DK)* includes all necessary circuitry, such as antennas and a *Subscriber Identity Module (SIM)* card holder, and provides developers access to all I/O pins and relevant module interfaces.

Note: The DK can be connected to a base station using the onboard LTE antenna or to an LTE emulator with an RF cable. See [nRF9160 certifications](#) for information on the bands supported by the LTE modem. The DK requires an *Universal Integrated Circuit Card (UICC)/SIM* for communication with the LTE network.

To get started with the nRF9160 DK, install [nRF Connect for Desktop](#) and from there install the Quick Start app.

Key features

- nRF9160 SiP
- Onboard LTE antenna which supports all bands supported by the SiP
- Onboard GNSS antenna
- nRF52840 as a board controller and network processor for *Bluetooth®* and IEEE 802.15.4 protocols
- Buttons, switches, and LEDs for user interaction
- I/O interface for Arduino form factor plug-in modules
- SEGGER J-Link OB Debugger with debug out functionality
- *Universal Asynchronous Receiver/Transmitter (UART)* interfaces through virtual serial ports
- *Universal Serial Bus (USB)* connection for debugging and programming and power
- SIM card socket for nano-SIM (4FF SIM)
- Interfaces for nRF9160 current consumption measurements

2 Kit content

The nRF9160 DK comes with a SIM card and an information leaflet.

In addition to the supplied hardware, the nRF9160 DK includes preprogrammed firmware, documentation, and hardware files.

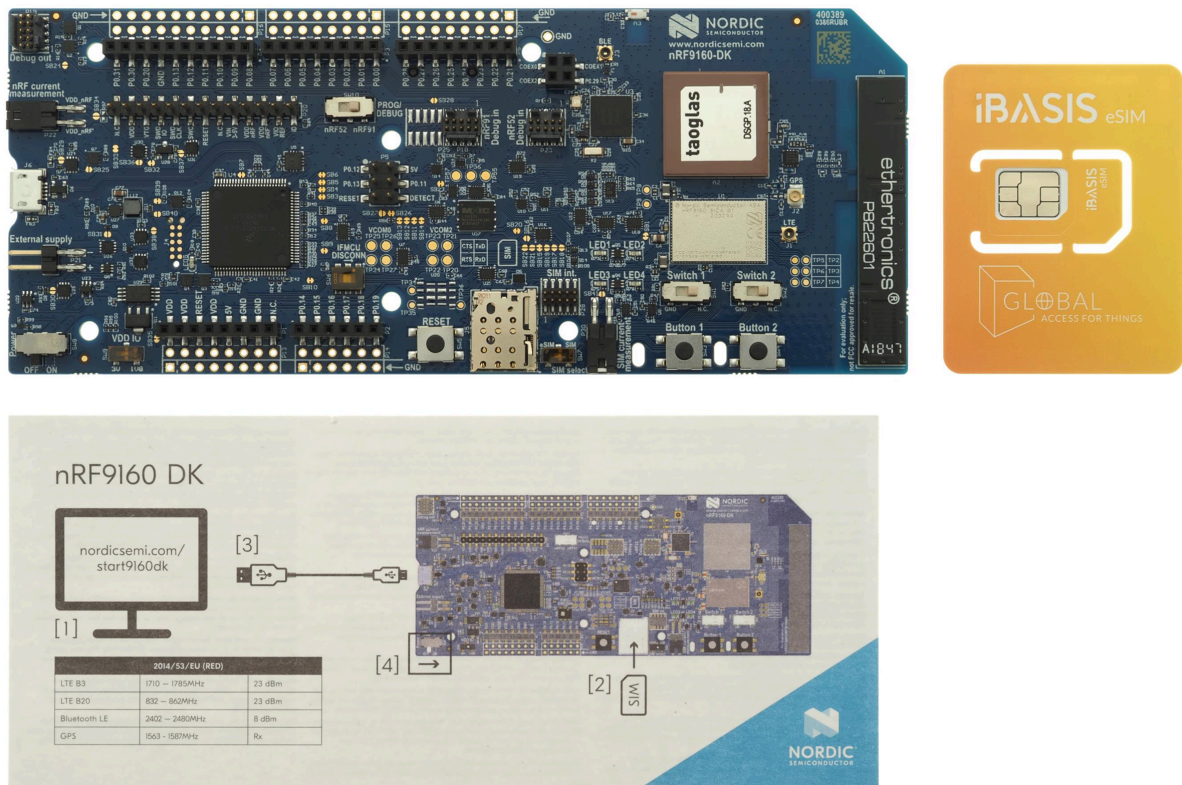


Figure 1: nRF9160 DK kit content

Hardware files

The hardware design files for the nRF9160 DK are available on the [nRF9160 DK](#) product page. They include the following resources:

- Schematics
- *Printed Circuit Board (PCB)* layout files
- Bill of materials
- Gerber files

3 Operating modes

The nRF9160 DK has two main modes of operation.

3.1 Firmware development mode

The primary interface for programming and debugging the nRF9160 DK is the **USB** port (**J6**). The USB port is connected to an interface MCU which embeds a **SEGGER J-Link-OB** (Onboard) debug probe.

The following figure shows the switches and interfaces used in firmware development mode. Firmware development mode is the default when the IFMCU DISCONN switch (**SW6**) is in the left position.

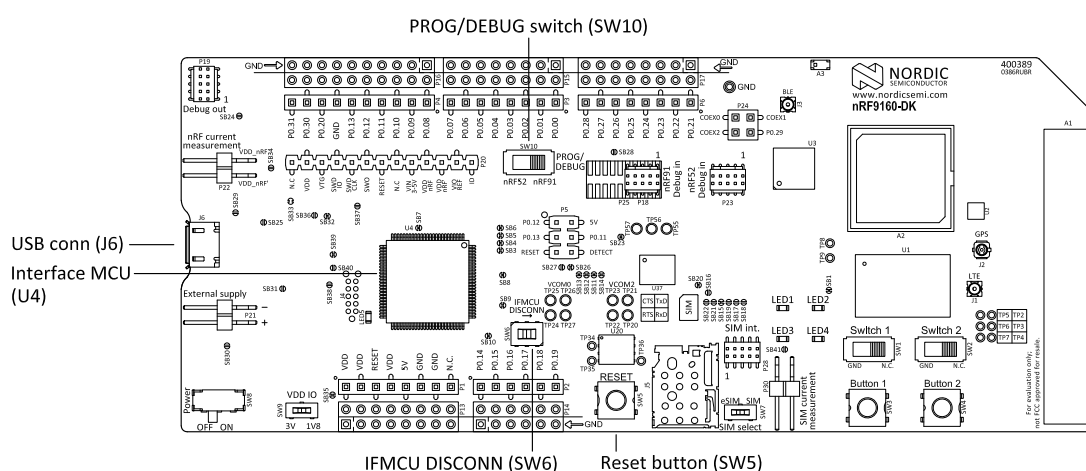


Figure 2: Firmware development mode

3.1.1 Device programming

The nRF9160 DK supports **Serial Wire Debug (SWD)** programming interfaces for onboard and off-board nRF targets.

The primary target for programming and debugging in the *DK* is the nRF9160 *SiP*. The interface MCU also supports programming the onboard nRF52840 as well as external nRF devices fitted on a shield or through a connector to external boards such as the user's own prototypes.

The interface MCU automatically detects connected external targets. The PROG/DEBUG switch (**SW10**) is used to select between the onboard nRF9160 and nRF52840 for programming or debugging. There are significant limitations on using the nRF52840 on the nRF9160 DK. For more information, see [Board control](#) on page 20.

3.1.2 Virtual serial port

The interface MCU features three **UART** interfaces through three virtual serial ports.

The virtual serial ports are the following:

- VCOM0 – Connected to nRF9160 *SiP* (default)
- VCOM1 – Connected to nRF52840 *System on Chip (SoC)* (nonconfigurable)
- VCOM2 – Connected to nRF9160 *SiP* (default)

For details on routing VCOM0 and VCOM2, see [Board control](#) on page 20.

The virtual serial ports have the following features:

- Flexible baud rate settings up to 1 Mbps
- *Request to Send (RTS)/Clear to Send (CTS)-style Hardware Flow Control (HWFC)*

The following table shows an overview of the *General-Purpose Input/Output (GPIO)*s used for the UART connections on nRF9160 and nRF52840:

Signal	nRF91 APP1	nRF91 APP2	nRF52840
TXD	P0 . 29	P0 . 01	P0 . 03
RXD	P0 . 28	P0 . 00	P0 . 05
CTS	P0 . 26	P0 . 15	P0 . 07
RTS	P0 . 27	P0 . 14	P1 . 08

Table 1: GPIOs used for virtual serial ports on nRF9160 and nRF52840

Note: Baud rate 921600 is not supported through the virtual serial port.

3.1.3 Mass storage device

The interface MCU features a *Mass Storage Device (MSD)*. This makes the DK appear as an external drive on your computer.

The MSD drive can be used for drag-and-drop programming. However, files cannot be stored on the drive. When a HEX file is copied to the drive, the interface MCU programs the file to the DK. The J-Link OB programs the target that is selected with the nRF52/nRF91 switch (**SW10**). For limitations on nRF52840 use, see [Board control](#) on page 20.

The following issues might occur during MSD operation:

- If Windows tries to defragment the MSD, the interface MCU disconnects and becomes unresponsive. To return to normal operation, power cycle the DK.
- The antivirus software might try to scan the MSD. Some antivirus programs trigger a false positive alert in one of the files and quarantine the unit. If this happens, the interface MCU becomes unresponsive.
- If the computer is set up to boot from USB, it can try to boot from the DK if it is connected. This is avoided by unplugging the DK before a computer restart or changing the boot sequence of the computer.

Disable the MSD of the kit by using the **msddisable** command in J-Link Commander. To enable the MSD, use the **msdenable** command. These commands take effect after a power cycle of the DK and stay this way until changed again.

3.1.4 Reset

The nRF9160 DK is equipped with a RESET button (**SW5**).

By default, the RESET button is connected to the interface MCU which forwards the reset signal to the nRF9160 SiP or the nRF52840 SoC depending on the state of the nRF52/nRF91 switch. If IF MCU DISCONNECT is activated, the RESET button is connected to the nRF9160 SiP.

3.2 Performance measurement mode

Select Performance Measurement mode by moving the IFMCU DISCONN switch (**SW6**) to the right position, which disconnects the interface MCU from the nRF9160 SiP using analog switches.

This isolates the nRF9160 SiP as much as possible and can be useful when measuring currents on low-power applications.

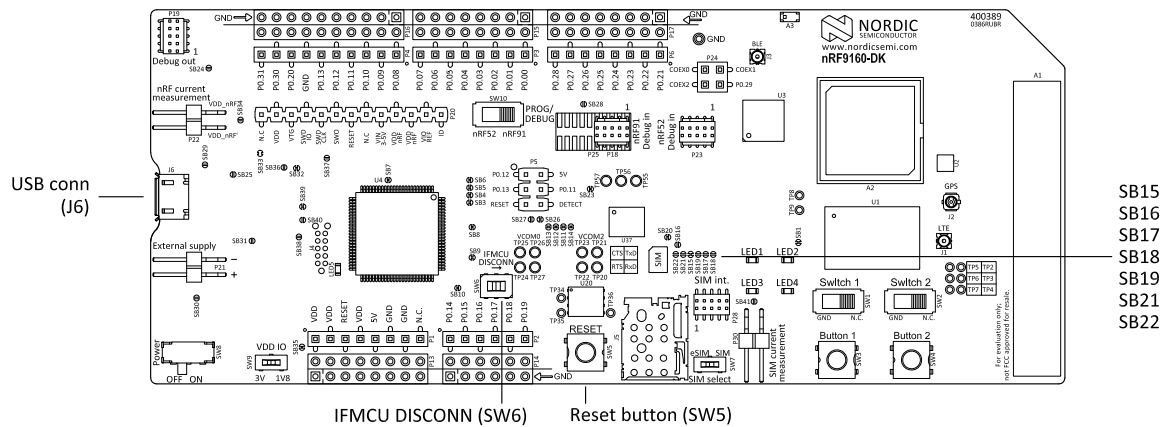


Figure 3: Performance measurement mode

The DK detects if a USB cable is plugged in (see [USB detect](#) on page 11) and disconnects and powers down the interface MCU when a supply other than the USB is used. If the USB connector is used for power supply only (USB battery back), the interface MCU can be disabled using **SW6**.

USB detect and **SW6** control the routing of the RESET signal from the RESET button (**SW5**). Normally, it is routed through the interface MCU, but if that is disabled, the reset button is routed directly to the nRF9160 SiP.

The following reset routing options are available through the use of solder bridges:

- When the interface MCU is disconnected, the RESET button is connected to **pin 32** (nRESET) of the nRF9160 SiP. The RESET button can be disconnected from the nRF9160 SiP by cutting **SB15**.
- When the interface MCU is disconnected, shorting **SB18** connects the RESET pin of the Arduino interface to the RESET pin of the nRF9160 SiP.
- When the interface MCU is connected, shorting **SB19** connects the RESET pin of the Arduino interface to the BOOT input of the interface MCU.
- Shorting **SB16** connects the RESET pin of the Arduino interface to the RESET button.
- Shorting **SB17** connects the RESET pin of the Arduino interface to the nRESET pin of the nRF9160 SiP.
- When the interface MCU is disconnected, shorting **SB21** connects the RESET pin of the nRF52840 to the RESET button.
- Shorting **SB22** connects the RESET pin of the nRF52840 to the RESET button.

3.2.1 USB detect

To detect when the interface MCU *USB* is connected, there is a circuit sensing the VBUS of the USB connector **J6**.

When the USB cable is connected, the VDD is propagated to the USB_DETECT signal.

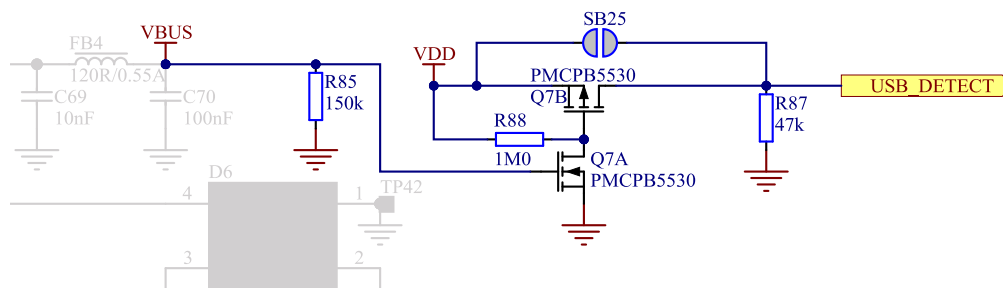


Figure 4: USB detect switch

4 Hardware description

The nRF9160 DK features an onboard programming and debugging solution.

4.1 Hardware drawings

The nRF9160 DK hardware drawings show both sides of the DK.

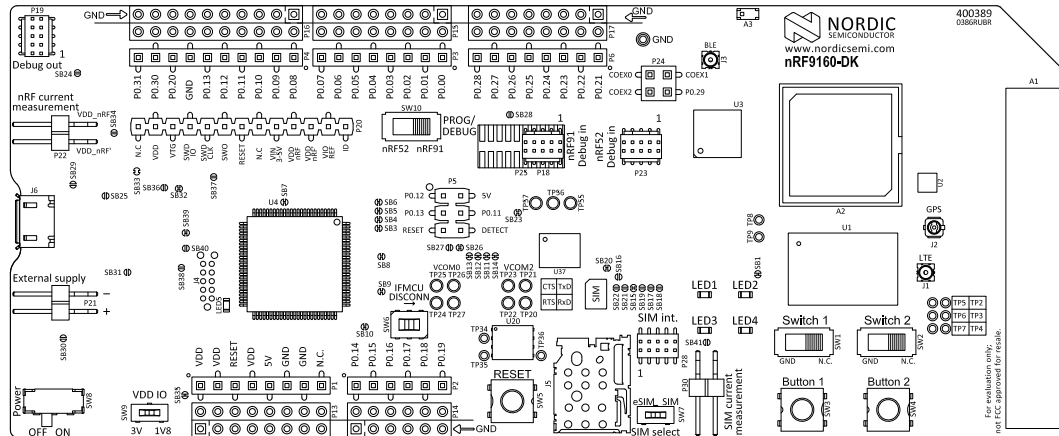
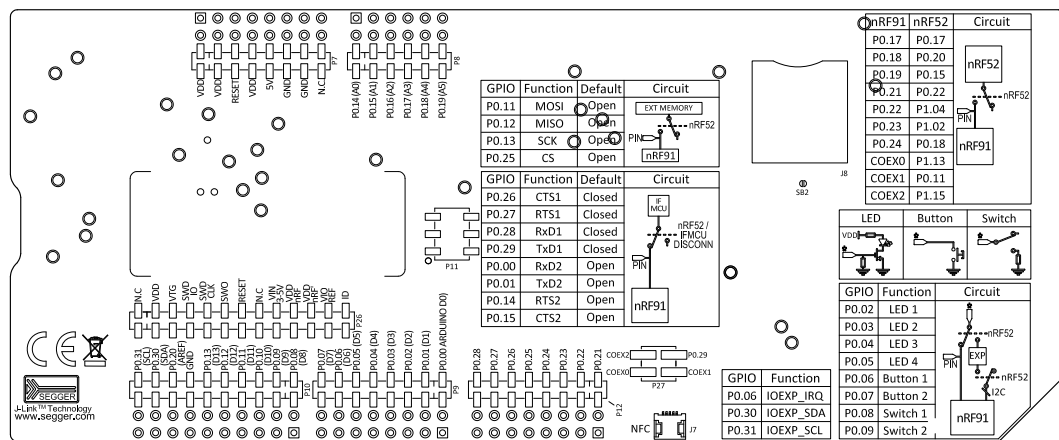


Figure 5: nRF9160 DK (PCA10090), front view



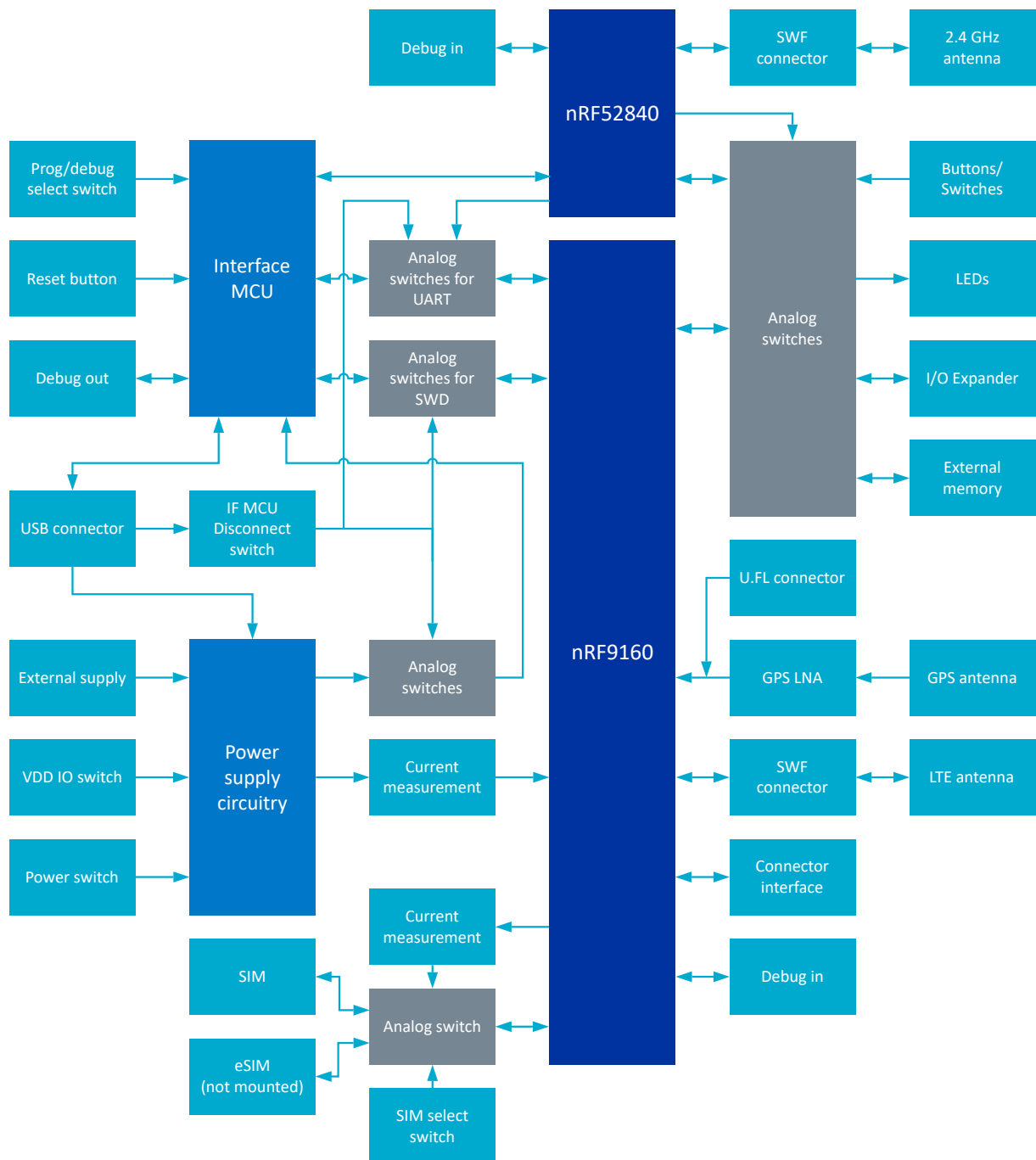


Figure 7: nRF9160 DK block diagram

4.3 Power supply

nRF9160 DK has a flexible and configurable power supply system to facilitate software development and testing using different power sources and accurate power measurements.

The power source options are the following:

- USB connector **J6** (5 V)
- External supply on **P21** (3.0 V to 5.5 V)
- VIN 3–5 V on **P20** (3.0 V to 5.5 V)

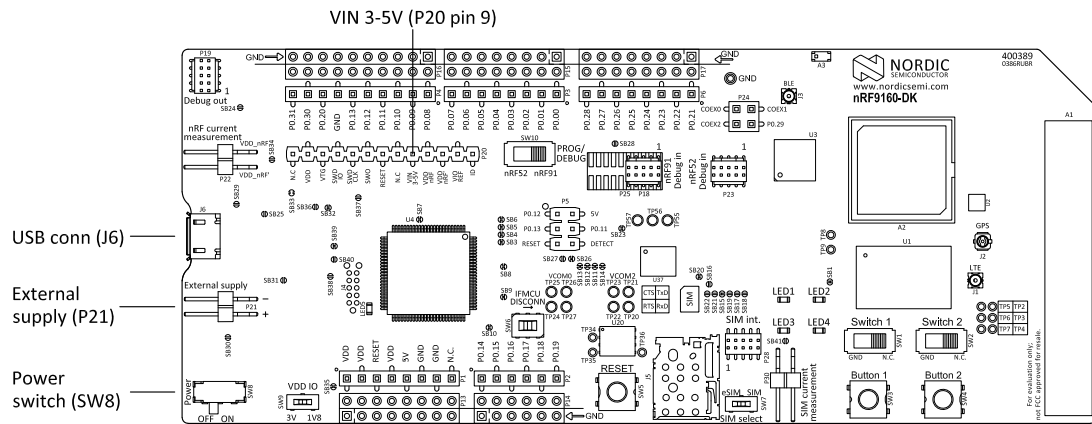


Figure 8: Power supply options

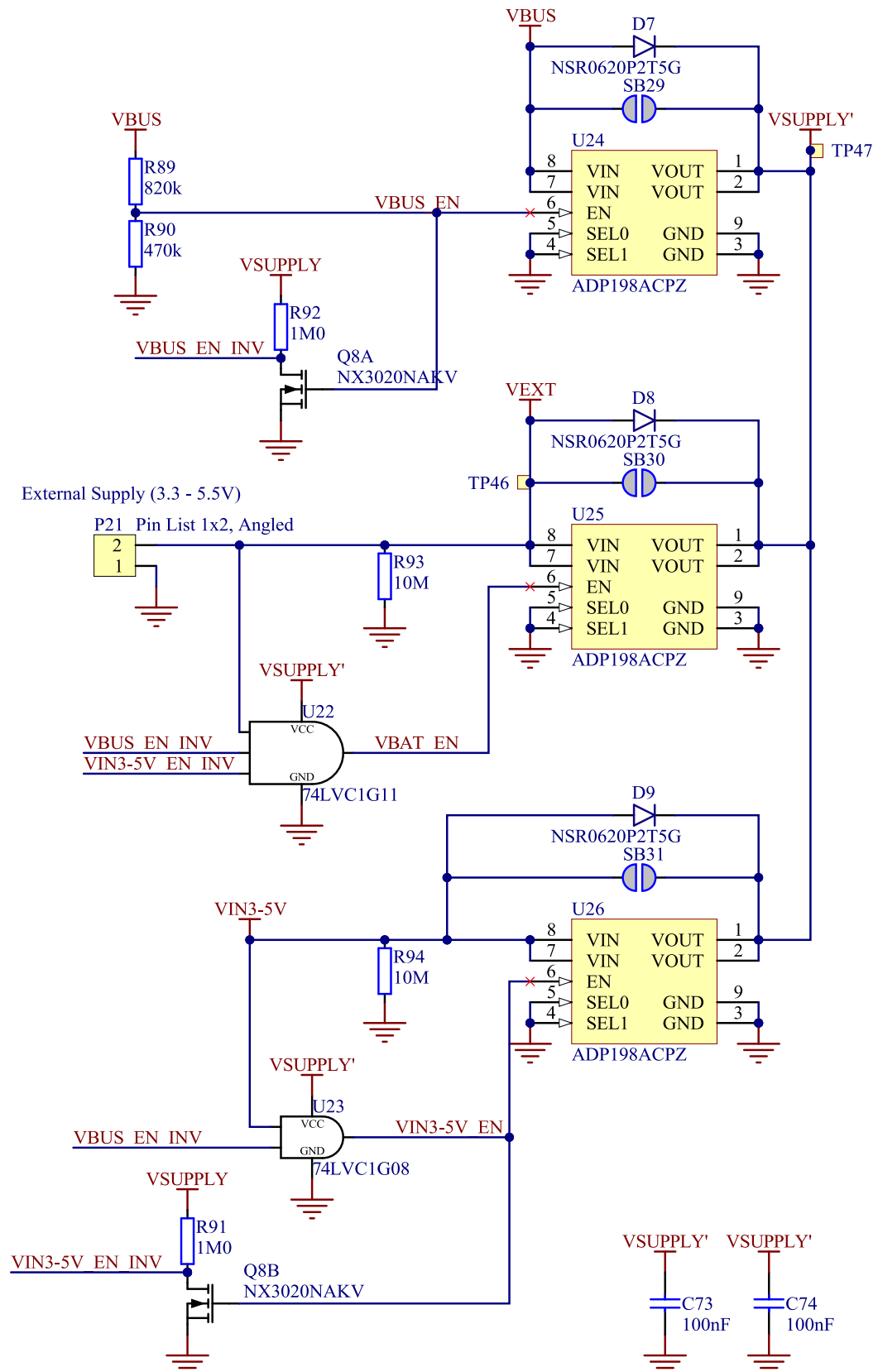


Figure 9: Power sources and switches

To ensure that only one power source is used on the DK at a time, power switches are implemented on each power source as shown in [Figure 9: Power sources and switches](#) on page 15. These switches prioritize the supply sources as follows:

1. **USB**
2. **P20**
3. **P21**

This means that if power is connected to more than one the interfaces, the higher priority interface is chosen to supply the DK.

The supply voltage is then routed through the ON/OFF switch (**SW8**) to the common rail VSUPPLY, which acts as the source for the supply voltage regulators for the circuitry on the DK.

The supply flows from VOUT to VIN. The body diode of the internal transistor powers the VSUPPLY net, which supplies the gates controlling the enable signal of the switches.

The power switches introduce a small voltage drop between the power source connected to the DK and the VSUPPLY. To avoid this, the power switches can be bypassed by shorting one of the solder bridges as shown in the following table:

Power source	Power switch bypass	Voltage level
USB connector J6	SB29	5 V
External supply P21	SB30	3.0 V to 5.5 V
VIN 3–5 V P20	SB31	3.0 V to 5.5 V

Table 2: Bypassing the power switch

CAUTION: To avoid damaging the DK, connect only one power source at a time in this case. Shorting the solder bridges removes the reverse voltage protection.

4.3.1 nRF9160 power supply

The nRF9160 has a supply range of 3.0 V to 5.5 V and is directly powered by the VSUPPLY rail.

4.3.2 VDD supply rail

VDD is the main supply for the rest of the circuitry on the nRF9160 DK. It is regulated down from VSUPPLY by a buck regulator (**U27**).

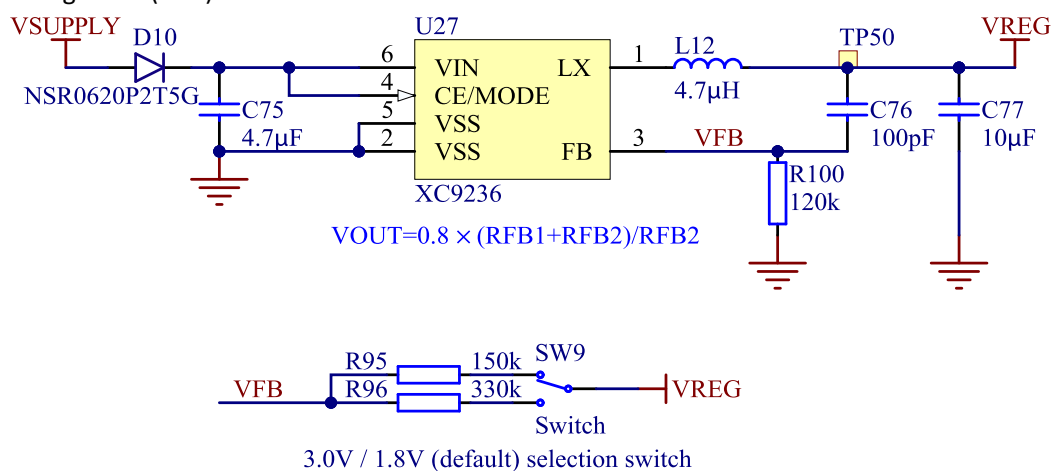


Figure 10: VDD buck regulator and selection switch

The VDD voltage can be set to 1.8 V (default) or 3 V with **SW9**. Running 3 V *GPIO* with heavy load may degrade the LTE RF performance. The VDD voltage level can also be customized by applying a custom

voltage level to VIO_REF (**P20**). For more information, see [GPIO - General purpose input/output](#) in the nRF9160 Product Specification.

VDD powers most of the other circuits and sets the GPIO signal amplitude between the nRF9160 SiP and other circuits on the DK including connectors and PIN headers.

4.3.3 Other power domains

The interface MCU needs 3.0 V for its *USB* interface supply. A low-dropout voltage regulator (**U28**) is used for this. The regulator also supplies the LEDs on the *DK* with a fixed supply.

4.4 Antenna interfaces

The nRF9160 DK has three antenna interfaces for *LTE*, *Global Navigation Satellite System (GNSS)*, and the 2.4 GHz radio.

The LTE and 2.4 GHz RF signals are propagated through two coaxial connectors with switches that can be used to disconnect the corresponding antenna from the radio if adapter cables are connected. This makes it possible to perform conducted measurements or to attach external antennas to the radio.

The GNSS signal is RX only. A *Low-Noise Amplifier (LNA)* with integrated filters amplifies and filters the signal before it is fed to the GNSS RF port on the nRF9160 DK. An external active GNSS antenna can be connected to **J2**. When using an external antenna, the LNA should be disabled.

The DK has the following antenna interface connectors:

- **J1** – Connector with a switch for the LTE antenna (**A1**)
- **J2** – Connector for an external GNSS antenna
- **J3** – Connector with a switch for the 2.4 GHz antenna (**A3**)

For more information on the GNSS antenna interface, see [GNSS](#) on page 17.

4.5 GNSS

The nRF9160 DK has a dedicated *GNSS* port to support global navigation. GNSS functionality requires support in the onboard network protocol firmware.

The GNSS signal is received from the onboard or external active GNSS antenna. The onboard antenna (**A2**) is connected to the *LNA* (**U2**).

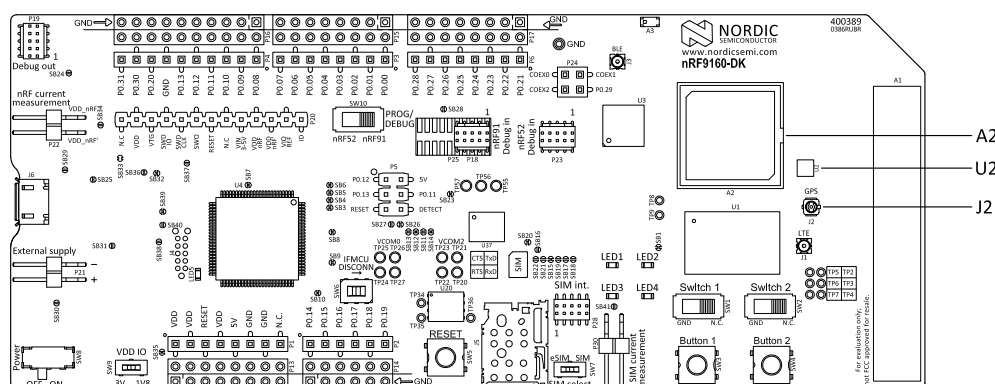


Figure 11: Onboard GNSS antenna (A2), LNA (U2), and connector J2

The GNSS signal from the onboard antenna is amplified and filtered in the LNA (**U2**), which has an integrated prefilter and postfilter, before it is fed to the nRF9160 DK. This makes the GNSS receiver more

sensitive to GNSS signals and less sensitive to interference from other sources on the *DK* or nearby. The LNA is enabled by a GNSS-enable signal from the nRF9160 DK using the **COEX0** pin.

An external active GNSS antenna can be connected to the *U.FL* connector **J2**. 3.3 V DC is fed through **J2** to supply the LNA of the external antenna. When using an external GNSS antenna, the onboard GNSS LNA (**U2**) must be disabled to avoid interference. This is done by ensuring that the GPS_EN signal is low and set by the **%XCOEX0 AT command**.

To optimize GNSS reception, do the following:

- GNSS signals do not usually penetrate ceilings or other structures that well. For best GNSS performance, the DK should be placed outside on a flat surface in an open space far from sources of interference and other structures that can block the satellite signals.
- The Molex patch antenna achieves the highest gain when placed horizontally on a surface (x-y) facing the z-axis since it can receive all propagated GNSS signals. A lower gain is experienced if the patch antenna is mounted at an angle.

The following figure shows the nRF9160 DK's GNSS circuitry.

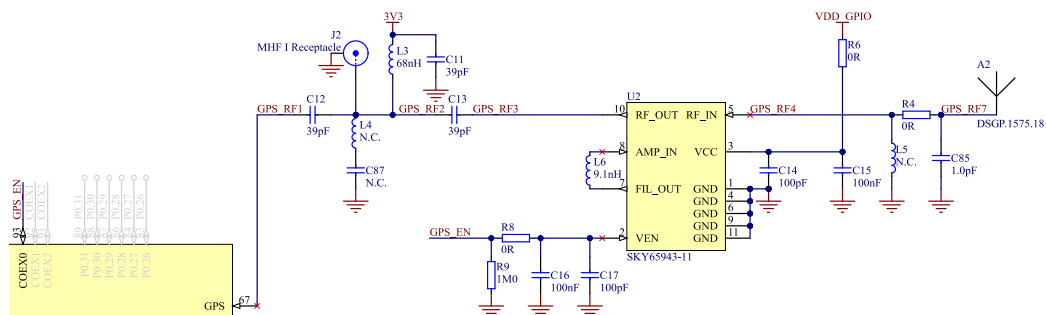


Figure 12: GNSS connected to the nRF9160

4.6 GPIO interfaces

Access to the nRF9160 *GPIOs* is available from connectors **P2**, **P3**, **P4**, **P6**, and **P24**. The nRF9160 DK supports the Arduino UNO interface.

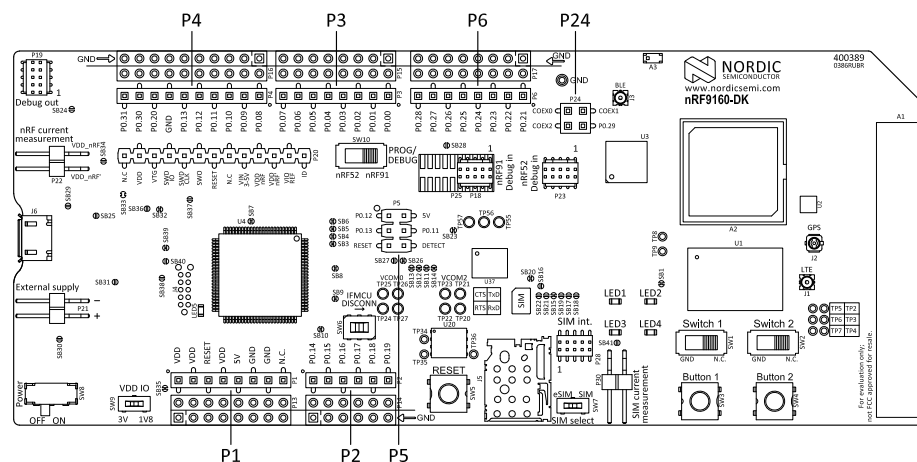


Figure 13: Access to GPIOs

GPIO signals are also available on connectors **P8**, **P9**, **P10**, **P12**, and **P27** which are on the back of the *DK*. By mounting pin headers on the connector footprints, the nRF9160 DK can be used as a shield for Arduino motherboards.

For easy access to GPIO, power, and ground, the signals can also be found on the through-hole connectors **P14**, **P15**, **P16**, and **P17**.

Note: GPIO P0.29 is not available on a through-hole connector.

The locations and default functions of the GPIO pins are described in the following table and figure.

GPIO nRF9160 DK	Function
P0.00, P0.01, P0.14, P0.15	Used as a second <i>UART</i> connection to the interface MCU. For more information, see Virtual serial port on page 9.
P0.02, P0.03, P0.04, P0.05, P0.06, P0.07, P0.08, P0.09	Connected to buttons, slide switches, and LEDs by default. For more information, see Buttons, slide switches, and LEDs on page 24.
P0.17, P0.18, P0.19, P0.21, P0.22, P0.23, P0.24, COEX0, COEX1, COEX2	Used to connect the nRF9160 SiP to the Board control on page 20.
P0.26, P0.27, P0.28, P0.29	Used as the primary <i>UART</i> connection to the interface MCU. For more information, see Virtual serial port on page 9.
P0.11, P0.12, P0.23, P0.25	Used for external memory interface. For more information, see External memory on page 27.

Table 3: Default pin settings

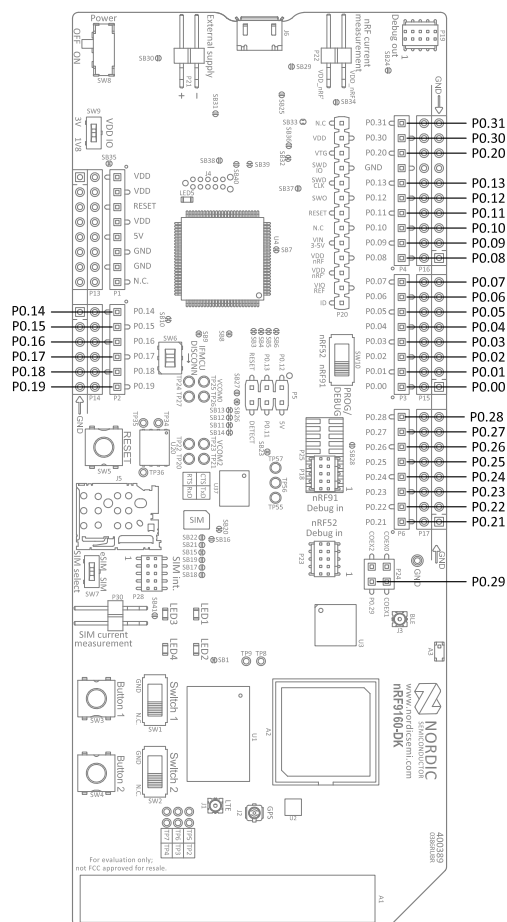


Figure 14: Pins

4.7 nRF52840

An nRF52840 Bluetooth/IEEE 802.15.4 SoC is included on the nRF9160 DK.

The nRF52840 SoC has the following functions:

- nRF9160 DK control
- Bluetooth/IEEE 802.15.4 network processor

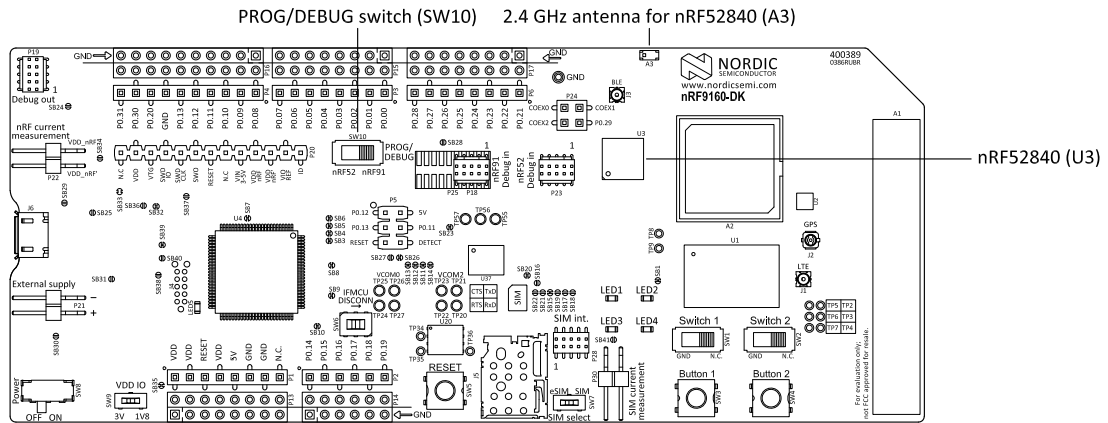


Figure 15: nRF52840 SoC on the nRF9160 DK

4.7.1 Board control

The nRF52840 SoC controls analog switches on the nRF9160 DK enabling routing of some of the nRF9160 GPIO pins to onboard functionality, for example LEDs, or the regular GPIO interfaces. The interface MCU controls the analog switches on the DK that define the board's operation.

The following table shows the GPIOs on the nRF9160 DK that can be routed by the analog switches.

Name	nRF52 control GPIO	Hardware default	nRF9160 GPIO	External ¹	Default	Optional
nRF91_UART1 (nRF91_APP1)	P1.14	0	P0.26	P6_pin6 (D19)	VCOM0_RTS	—
			P0.27	P6_pin7 (D20)	VCOM0_CTS	—
			P0.28	P6_pin8 (D21)	VCOM0_TXD	—
			P0.29	P24_pin4 (D22)	VCOM0_RXD	—
nRF91_UART2 (nRF91_APP2)	P1.12	1	P0.01	P3_pin2 (D1)	VCOM2_RXD	—
			P0.00	P3_pin1 (D0)	VCOM2_TXD	—
			P0.15/ AIN2	P2_pin2 (A1)	VCOM2_RTS	—
			P0.14/ AIN1	P2_pin1 (A0)	VCOM2_CTS	—
nRF91_LED1	P1.05	0	P0.02	P3_pin3 (D2)	LED 1	—
nRF91_LED2	P1.07	0	P0.03	P3_pin4 (D3)	LED 2	—
nRF91_LED3	P1.01	0	P0.04	P3_pin5 (D4)	LED 3	—
nRF91_LED4	P1.03	0	P0.05	P3_pin6 (D5)	LED 4	—

Name	nRF52 control GPIO	Hardware default	nRF9160 GPIO	External ¹	Default	Optional
nRF91_SWITCH1	P1.09	0	P0.08	P3_pin7 (D6)	Switch 1	–
nRF91_SWITCH2	P0.08	0	P0.09	P3_pin8 (D7)	Switch 2	–
nRF91_BUTTON1	P0.06	0	P0.06 ²	P4_pin1 (D8)	Button 1	–
nRF91_BUTTON2	P0.26	0	P0.07	P4_pin2 (D9)	Button 2	–
nRF_IF0-2_CTRL (nRF91_GPIO)	P0.13	0	P0.17/ AIN4	P2_pin4 (A3)	–	nRF52_P0.17
			P0.18/ AIN5	P2_pin5 (A4)	–	nRF52_P0.20
			P0.19/ AIN6	P2_pin6 (A5)	–	nRF52_P0.15
nRF_IF3-5_CTRL (nRF91_TRACE)	P0.24	0	P0.21	P6_pin1 (D14) / P25_pin12 (TRACECLK)	–	nRF52_P0.22
			P0.22	P6_pin2 (D15) / P25_pin14 (TRACEDATA0) ³	–	nRF52_P1.04
			P0.23	P6_pin3 (D16) / P25_pin16 (TRACEDATA1) ³	–	nRF52_P1.02
nRF_IF6-8_CTRL (nRF91_COEX)	P1.10	0	GPS_EN	P24_pin1 (COEX0)	–	nRF52_P1.13
			COEX1	P24_pin2 (COEX1)	–	nRF52_P1.11
			COEX2	P24_pin3 (COEX2)	–	nRF52_P1.15
nRF_IF9_CTRL	P0.16	0	P0.24	P6_pin4 (D17) / P25_pin28 (TRACEDATA2) ³	–	nRF52_P0.18_Reset
IO_EXP_EN	P0.14	0	P0.06 ²	P3_pin7 (D6)	–	IO_EXP nINT
			P0.30	P4_pin9 (SDA)	–	IO_EXP SDA
			P0.31	P4_pin10 (SCL)	–	IO_EXP SCL
EXT_MEM_CTRL	P0.19	0	P0.11 (MOSI)	P4_pin4 (D11)	–	EXT_FLASH SI
			P0.12 (MISO)	P4_pin5 (D12)	–	EXT_FLASH SO
			P0.13/ AIN0 (SCK)	P4_pin6 (D13)	–	EXT_FLASH CSK

Name	nRF52 control GPIO	Hardware default	nRF9160 GPIO	External ¹	Default	Optional
			P0.25 (nCS)	P6_pin5 (D18) / P25_pin20 (TRACEDATA3) ^{3,4}	–	EXT_FLASH nCS

¹For the location of connectors, see [Figure 5: nRF9160 DK \(PCA10090\), front view](#) on page 12.

²When the I/O expander is enabled, the interrupt line from the I/O expander shares the same GPIO as Button 1.

³Pins are shared with TRACE. When TRACE functionality is to be used, make sure that the nRF52 control GPIO is in **Hardware default** state.

⁴Solder bridge and test point are available and so functionality can be maintained by manually connecting the signal to a GPIO of choice even when TRACE is in use.

Table 4: Board control routing

Name

The name to be used when referring to this signal path and the configuration needed to control it.

nRF52 control GPIO

The GPIOs on nRF52840 that are used to control the analog switches.

Hardware default

The default logic level of the control signal if pin is not driven.

nRF9160 GPIO

The GPIOs on nRF9160 that can be configured.

External

The connector and pin with permanent routing of the nRF9160 GPIO, not affected by the board controller.

Default

The default connection for the nRF9160 GPIO if the nRF52 control GPIO pins are not driven (Hardware default logic level).

Optional

The optional connection for the nRF9160 GPIO if the nRF52 control GPIO pins are driven to the opposite logic level than the hardware default.

The setup described as Default in [Table 4: Board control routing](#) on page 20 is active if all the GPIOs on nRF52840 listed as nRF52 control GPIO are not driven or are driven according to the hardware default setting. There are pull resistors on the switch control lines, and therefore, Default can also be received if the control lines from the nRF52840 are not driven. If the optional routing is required, the nRF52840 control GPIO pins must be set as outputs and driven high.

The nRF52840 SoC is preprogrammed with firmware that provides a control setting. This might not be the same as the hardware default. Configurable code examples for the nRF52840 SoC allowing to change the board routing can be found in the nRF Connect SDK. After the changes are done, the new program must be compiled and programmed to the nRF52840. See the nRF Connect SDK documentation on [nRF9160 board controller](#) for information on changing the configuration.

Note: To program and debug the nRF52840 SoC, nRF52 must be selected on the **PROG/DEBUG SW10** switch (**SW5** for v0.9.0 and earlier).

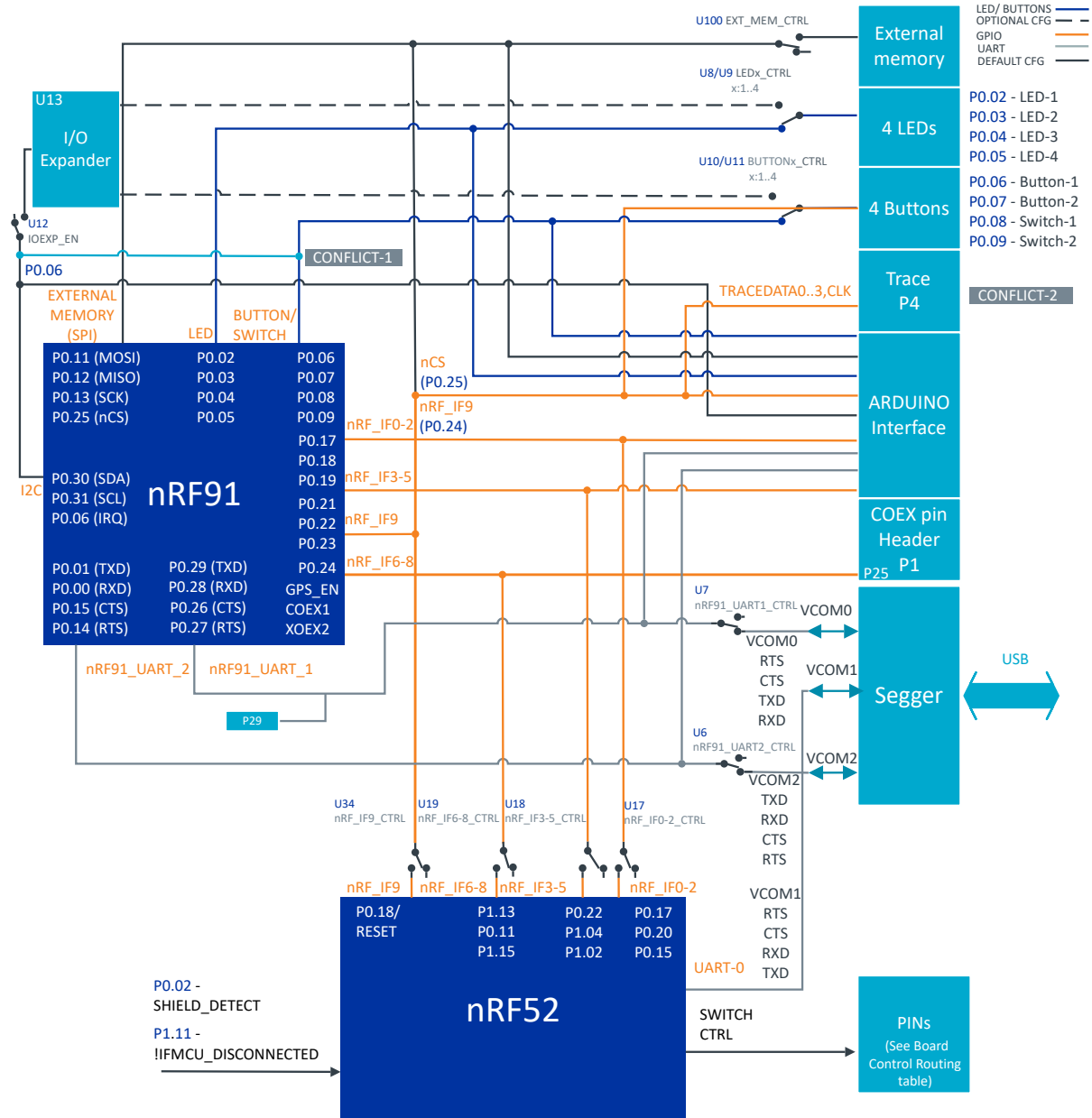


Figure 16: nRF9160 DK board control

Since this firmware in the nRF52840 SoC determines the nRF9160 DK behavior, it is vital that it is always present in the nRF52840 SoC. If it is accidentally erased or firmware affecting the use of the key nRF52840 GPIOs is programmed in, nRF9160 DK functionality is not guaranteed. In such cases, nRF52840 must be preprogrammed with the default board control firmware from the nRF Connect SDK.

4.7.2 Bluetooth/IEEE 802.15.4 network processor

The onboard nRF52840 SoC acts as a Bluetooth or IEEE 802.15.4 network processor. This supports development of products such as a cellular to Bluetooth and/or IEEE 802.15.4 bridge or gateway.

Since most of the *GPIOs* of the nRF52840 SoC are used for board control, it is not intended to be a full programming and development target on the nRF9160 DK.

However, when combined with board control firmware, it can run code that requires limited access to GPIO, such as wireless controllers or network processor implementations of Bluetooth and IEEE 802.15.4 protocols from Nordic Semiconductor.

This enables the nRF9160 DK to implement a cellular to Bluetooth (and/or IEEE 802.15.4) bridge or gateway. To enable this, nRF52840 has a 2.4 GHz antenna fitted along with nine GPIOs that can be connected to nRF9160 through the board control routing. For details, see [Table 4: Board control routing](#) on page 20.

Firmware examples for such bridge or gateway implementations can be found in [Cellular: LTE Sensor Gateway](#).

The following figure shows nRF52840 GPIOs and circuitry.

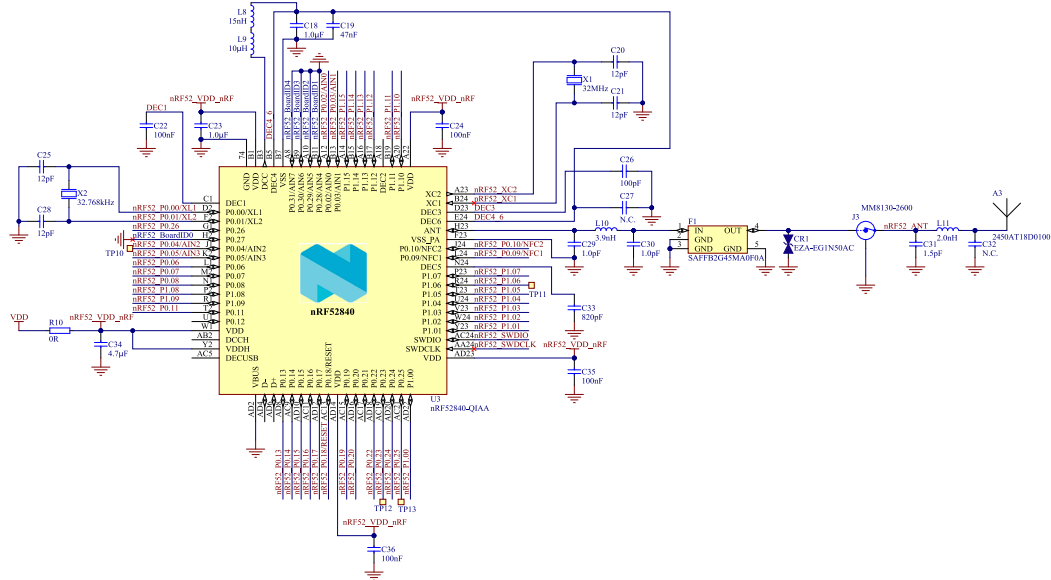


Figure 17: nRF52840 board controller

4.8 Buttons, slide switches, and LEDs

The nRF9160 DK has four LEDs, two buttons, and two switches for user interaction. By default, they are connected to the nRF9160 GPIOs as shown in the following table.

Part	GPIO
LED 1	P0.02
LED 2	P0.03
LED 3	P0.04
LED 4	P0.05
Button 1	P0.06
Button 2	P0.07
Switch 1	P0.08
Switch 2	P0.09

Table 5: Button and LED connection

To change default nRF9160 GPIO connections, see [Board control](#) on page 20

The nRF52840 SoC can be used to control the connection of any of the LEDs, slide switches, or buttons. Any LED, slide switch, or button can optionally be routed to an I/O expander. For more information, see [I/O expander](#) on page 25.

The buttons and switches are active low, meaning that the input is connected to ground when the buttons are pushed or the switches are slid to the GND position. The buttons and switches have no external pull-up resistor, and therefore the **P0.06**, **P0.07**, **P0.08**, and **P0.09** pins must be configured as an input with an internal pull-up resistor.

The LEDs are active high, meaning that writing a logical one (1) to the output pin illuminates the LED. The nRF9160 GPIOs control power transistors, and the LEDs are fed from a separate 3.0 V domain. Therefore, LED current is not drawn from nRF9160 GPIOs or the nRF9160 supply.

The following figures show the buttons, switches, and LEDs.

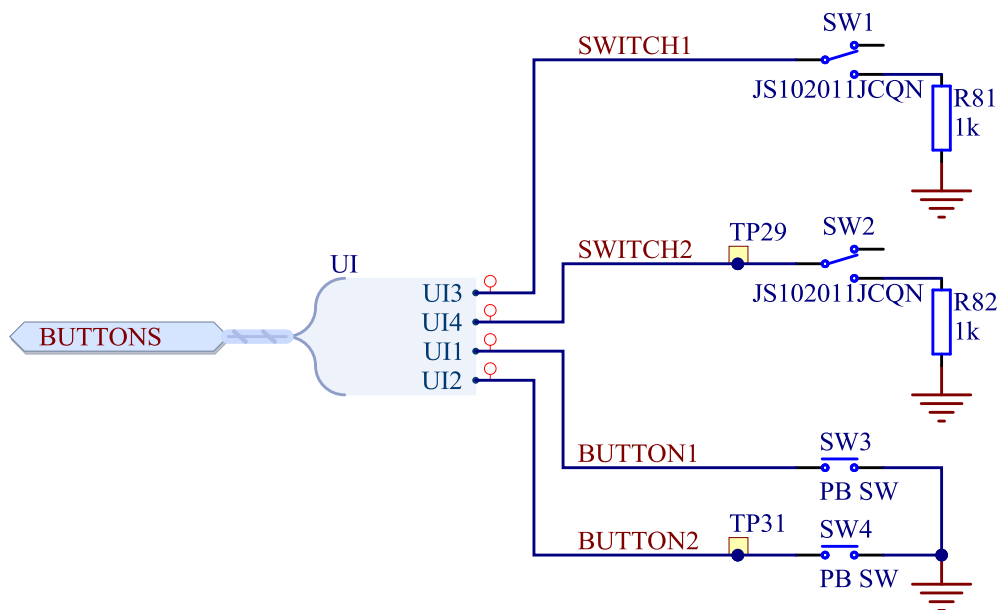


Figure 18: Buttons and switches

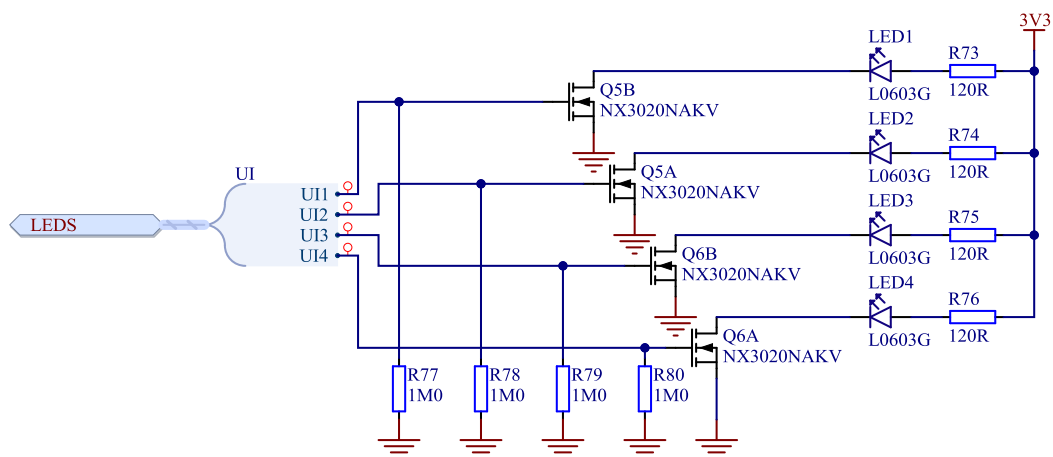


Figure 19: LEDs

4.8.1 I/O expander

The nRF9160 DK has an I/O expander that can optionally be used to interface the LEDs, slide switches, and buttons.

The connection of each component is controlled by the nRF52840 SoC using analog switches (see [Switches for buttons and LEDs](#) on page 34). The I/O expander is enabled by the nRF52840 SoC (see [Switches for I/O expander](#) on page 37). When the I/O expander is enabled, the I/O expander is interfaced by the nRF9160 DK *GPIOs* shown in the following table.

Signal	GPIO	Description
SDA	P0.30	I2C data line
SCL	P0.31	I2C clock line
IOEXP_IRQ	P0.06	Interrupt line from the I/O expander

Table 6: I/O expander interface

The following table shows the I/O expander connections.

Component	I/O expander pin
LED 1	I/O_EXP_IO4
LED 2	I/O_EXP_IO5
LED 3	I/O_EXP_IO6
LED 4	I/O_EXP_IO7
Switch 1	I/O_EXP_IO2
Switch 2	I/O_EXP_IO3
Button 1	I/O_EXP_IO0
Button 2	I/O_EXP_IO1

Table 7: I/O expander connections

The following figure shows the I/O expander connections.

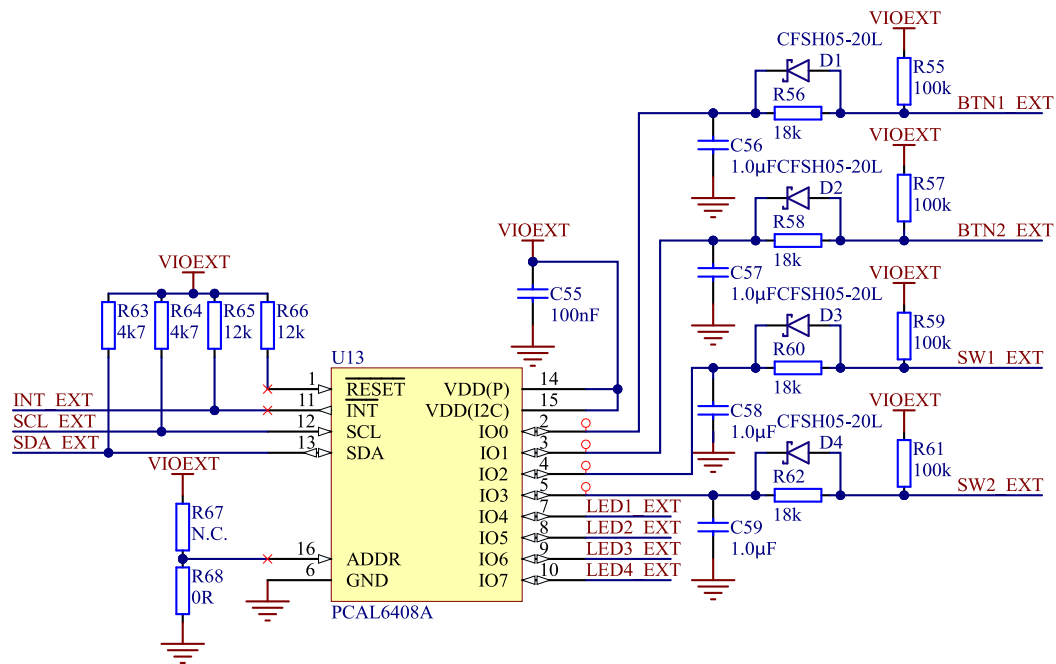


Figure 20: I/O expander

4.9 External memory

The nRF9160 DK has a 64 Mb external flash memory. The memory is a multi-I/O memory supporting both SPIs.

The following table shows flash memory GPIO usage, solder bridges, and test points.

GPIO	Flash memory pin	Solder bridge for default use	Test point
P0.25	CS	SB23	TP57
P0.13	SCLK	N/A	N/A
P0.11	SIO_0/SI	N/A	N/A
P0.12	SIO_1/SO	N/A	N/A
N/A	SIO_2/WP	N/A	TP55
N/A	SIO_3/HOLD	N/A	TP56

Table 8: Flash memory GPIO usage, solder bridges, and test points

The flash memory CS pin is by default connected to P0.25, which is also used for trace. To use the external memory together with trace, cut solder bridge SB23 and strap a wire from test point TP57 to a GPIO of choice. The WP and HOLD pins of the external memory are available only on test points TP55 and TP56, and to use these, strap wires from the test points to the GPIOs of choice.

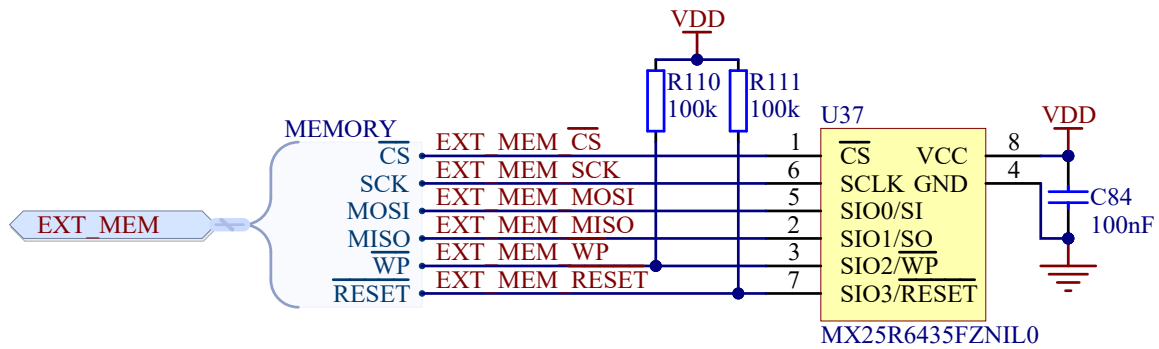


Figure 21: Flash memory

4.10 Debug input and trace options

The primary debug interface on the nRF9160 DK is the Segger OB debugger available through the *USB* port. If a power supply other than USB is used on the *DK*, this functionality is disabled.

The Debug in connectors **P18** (nRF9160) and **P23** (nRF52840) make it possible to connect external debuggers for debugging when the interface USB cable is not connected or if the DK is in IF MCU DISCONNECT mode.

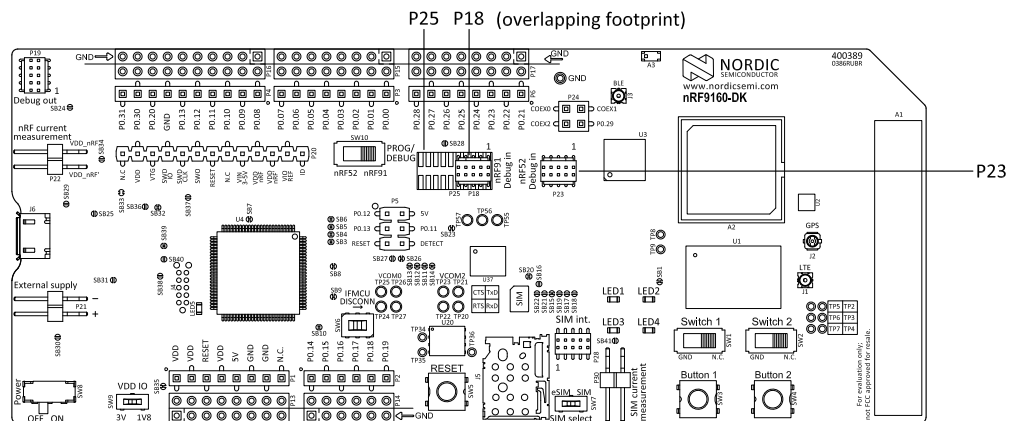


Figure 22: Debug input connector and trace footprint

To utilize the software trace feature on nRF9160, a footprint for a 20-pin connector (**P25**) is available. If trace functionality is required, a 2×10 pin 1.27 mm pitch surface mount connector can be mounted. nRF9160 *GPIOs* used for the trace interface are not available for application firmware use during trace.

Note: Connectors **P18** and **P25** overlap and share the same footprint.

GPIO	Trace
P0.21	TRACECLK
P0.22	TRACEDATA[0]
P0.23	TRACEDATA[1]
P0.24	TRACEDATA[2]
P0.25	TRACEDATA[3]

Table 9: Trace interfaces

4.11 Debug out for programming external boards

The nRF9160 DK supports programming and debugging external boards with nRF51 Series, nRF52 Series, and nRF53 Series SoCs and nRF91 Series SiPs.

The interface MCU on the nRF9160 DK runs SEGGER J-Link OB Debugger interface firmware and is used to program and debug the firmware of the nRF9160 SiP by default.

To program or debug an external board, connect to the Debug out connector (**P19**) using a 10-pin cable or use **P20** for custom connection.

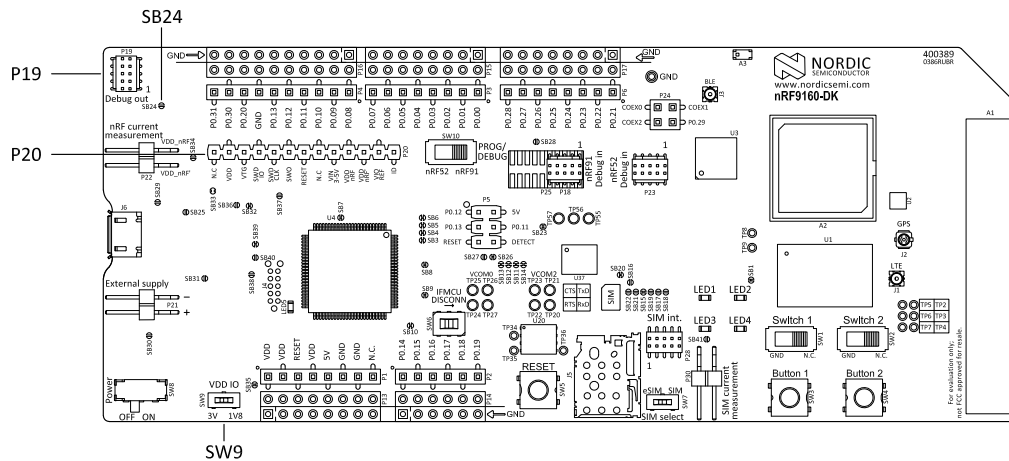


Figure 23: Debug output connectors

4.11.1 Programming an external board

For boards with a standard 10-pin SWD connector or a connector that supports a standard 10-pin flat cable, connection to **P19** is recommended.

Connect the boards as shown in the following figure.

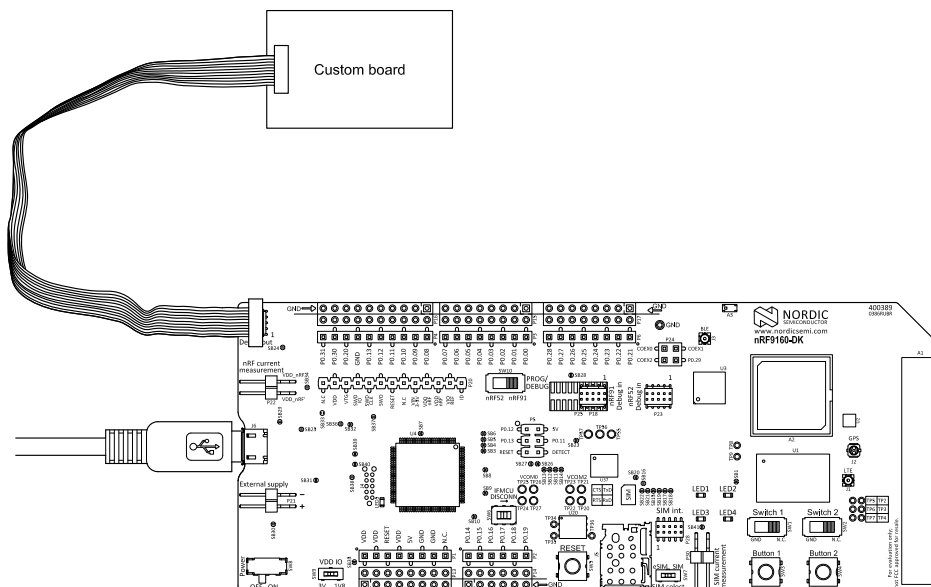


Figure 24: Connecting an external board to P19

Note: It is recommended to power the external board separately from the DK. The voltage on the external board must match that of the DK, which is 1.8 V or 3.0 V depending on the position of **SW9** when the DK is powered through the USB connector.

It is recommended to power the external board separately from the DK. The voltage of the SWD interface on the external board must match that of the DK, which is 1.8 V or 3.0 V depending on the position of **SW9** when the DK is powered through the USB connector.

When the interface MCU detects the voltage of the external board on **pin 1 (VTG)** of **P19**, it programs or debugs the target chip on the external board instead of the onboard nRF9160 SiP.

The following figure and table show the **P19** pinout.

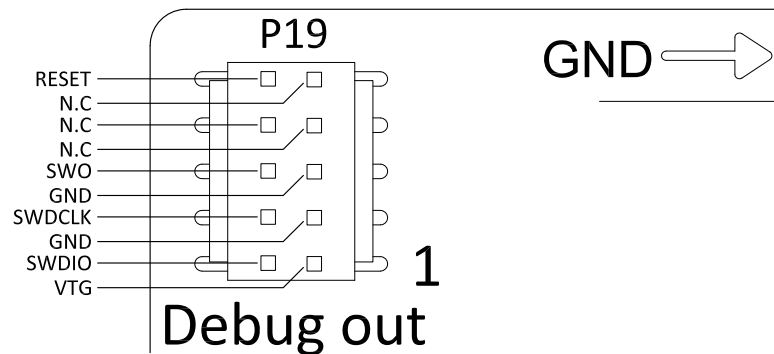


Figure 25: Debug output connector P19

Pin number	Signal	Description
1	VTG	Voltage supply from the external target, used as a select pin for the interface MCU to enable this port
2	SWDIO	SWD data input/output
3	GND	Ground
4	SWDCLK	Serial wire clock line
5	GND	Ground
6	SWO	Serial Wire Output (SWO) line is not used for programming and debugging over SWD
7	N.C.	Not used
8	N.C.	Not used
9	N.C.	Not used
10	RESET	Reset line

Table 10: Pinout of connector P19 for programming external targets

If you do not have a separate power supply on the external board connected to the Debug out connector **P19**, the external board can be powered from the onboard DK regulator as follows:

1. Short solder bridge **SB24** to enable power output on **P19**.
2. Power the DK through the USB connector.

CAUTION: To avoid damaging your board when **SB24** is shorted, do not connect a separate power supply to the external board and follow the instructions carefully.

Note: When **SB24** is shorted, it is not possible to program the onboard nRF9160 SiP or the nRF52840 SoC or to use the debug output on **P20** even if the external board is unplugged.

4.11.2 Programming a board with custom connections

If your external board has custom connections for programming and debugging pins, you can use the debug output on **P20**.

Connect the boards as shown in the following figure.

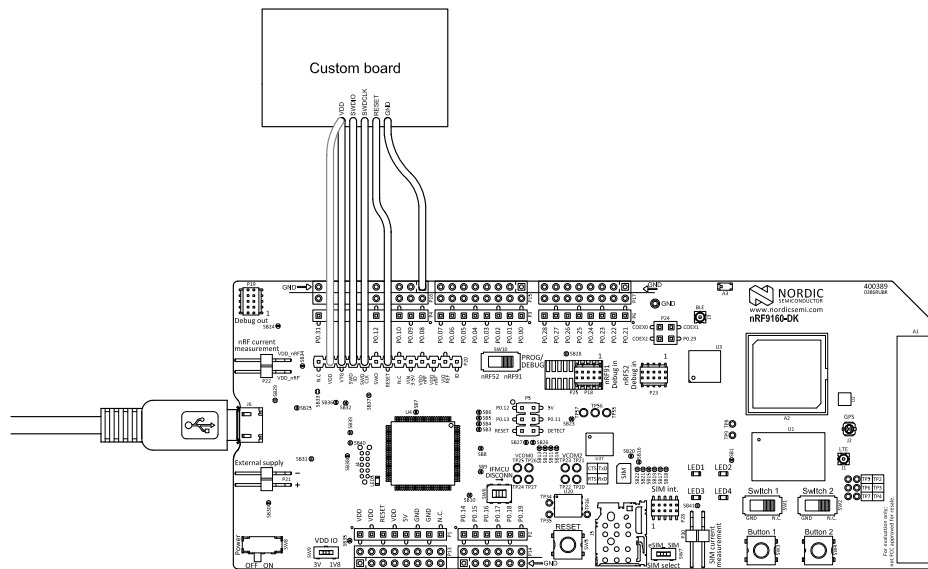


Figure 26: Connecting a custom board to P20

It is recommended to power the external board separately from the DK. The voltage of the SWD interface on the external board must match that of the DK, which is 1.8 V or 3.0 V depending on the position of **SW9** when the DK is powered through the USB connector.

When the interface MCU detects the voltage of the external board on **pin 3 (SELECT) of P20**, it programs or debugs the target chip on the external board instead of the onboard nRF9160 SiP.

Note: If the interface MCU detects power on **P19** and **P20**, it programs or debugs the target connected to **P19** by default.

If there is no separate power supply on the external board, the nRF9160 DK can supply power through pin 2 (**VDD**) of **P20**.

CAUTION: To avoid damaging your board, when **VDD** of nRF9160 DK is connected to the external board, do not connect a separate power supply to the external board.

The following figure and table show the **P20** connector pinouts.

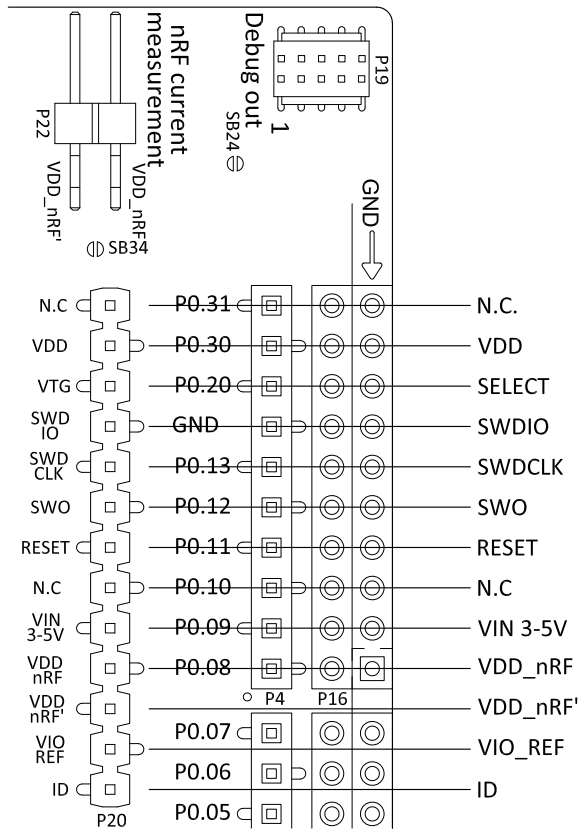


Figure 27: Debug output connector P20

Pin number	Signal	Description
1	N.C.	Not used
2	VDD	Main nRF9160 DK <i>GPIO</i> level power domain
3	VTG	Voltage supply from the external target, used as a select pin for the interface MCU to enable this port
4	SWDIO	<i>SWD</i> data line
5	SWDCLK	Serial wire clock line
6	SWO	The <i>SWO</i> line is not used for programming and debugging over <i>SWD</i>
7	RESET	Reset line
8	N.C.	Not used
9	VIN 3-5V	Voltage supply
10	VDD_nRF	nRF9160 SiP power domain
11	VDD_nRF'	Main nRF9160 DK high voltage power domain
12	VIO_REF	GPIO voltage reference input
13	ID	ID resistor to ground

Table 11: Pinout of connector P20 for programming external targets

4.12 Signal routing switches

Several of the *GPIO* signals of the nRF9160 DK are routed through analog switches for onboard functionality or for having them available on the pin headers for external circuitry or Arduino-type shields.

4.12.1 Interface MCU disconnect switches

The IF MCU DISCONNECT mode can be enabled using the following switches.

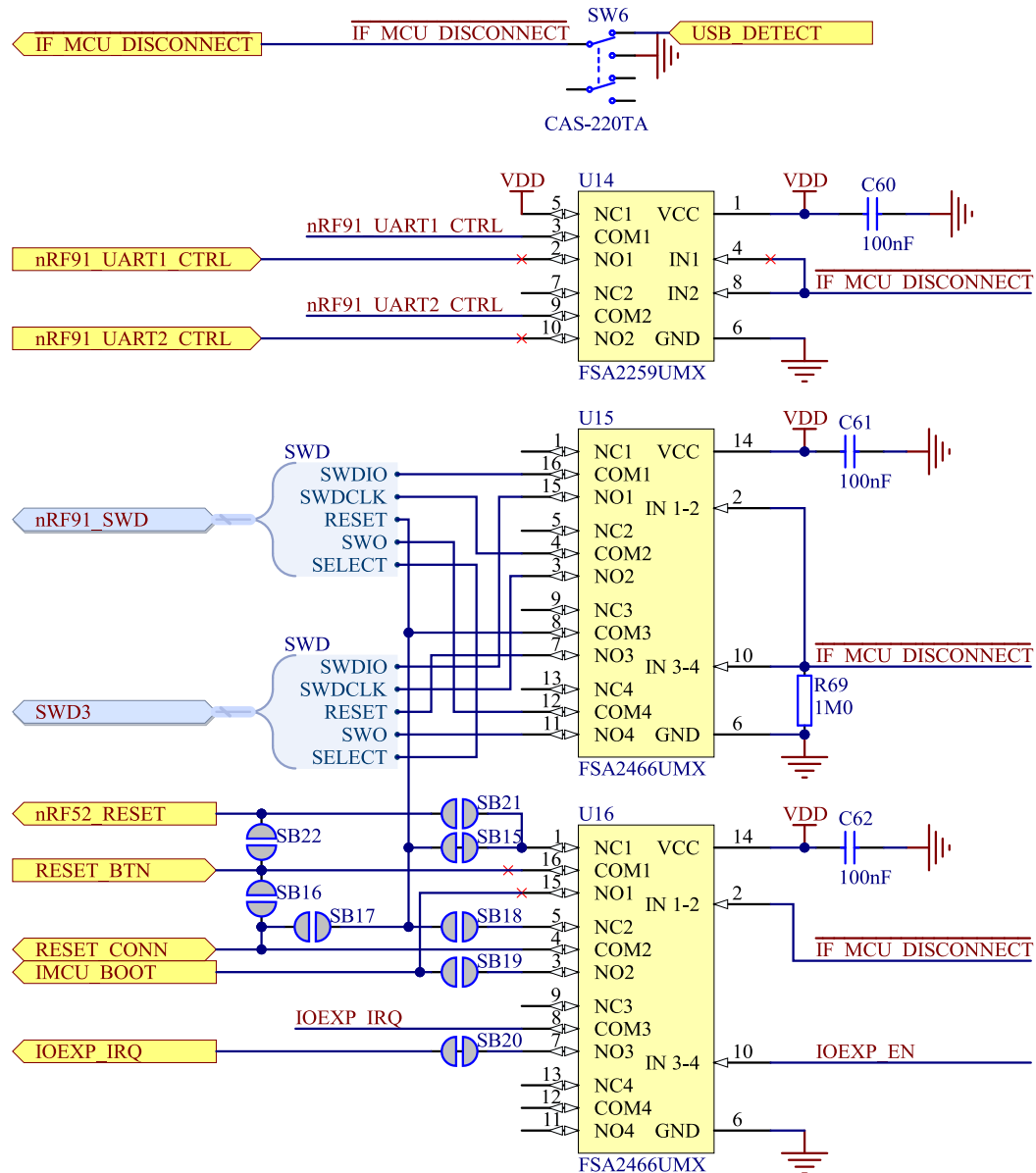


Figure 28: Signal interface switches

For more information, see [Performance measurement mode](#) on page 10.

4.12.2 Switches for UART interface

Two UART interfaces are routed between the interface MCU and the nRF9160 SiP. These can be controlled individually.

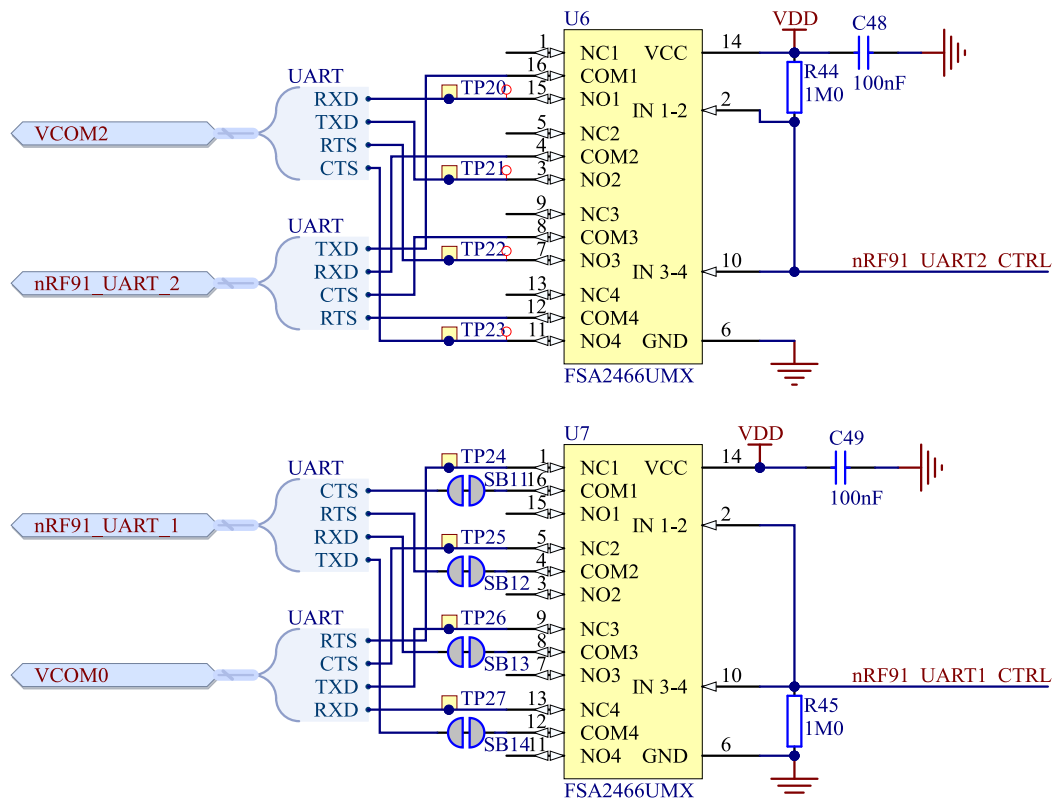


Figure 29: UART interface switches

4.12.3 Switches for buttons and LEDs

On the nRF9160 DK, there are analog switches that are used to connect and disconnect signals to control buttons, switches, and LEDs.

The analog switches control whether the LEDs, buttons, and switches are connected directly to the nRF9160 *GPIO* or to an I/O expander. The switches are controlled by the nRF52840 SoC, which contains Nordic Semiconductor firmware. For more information, see [I/O expander](#) on page 25.

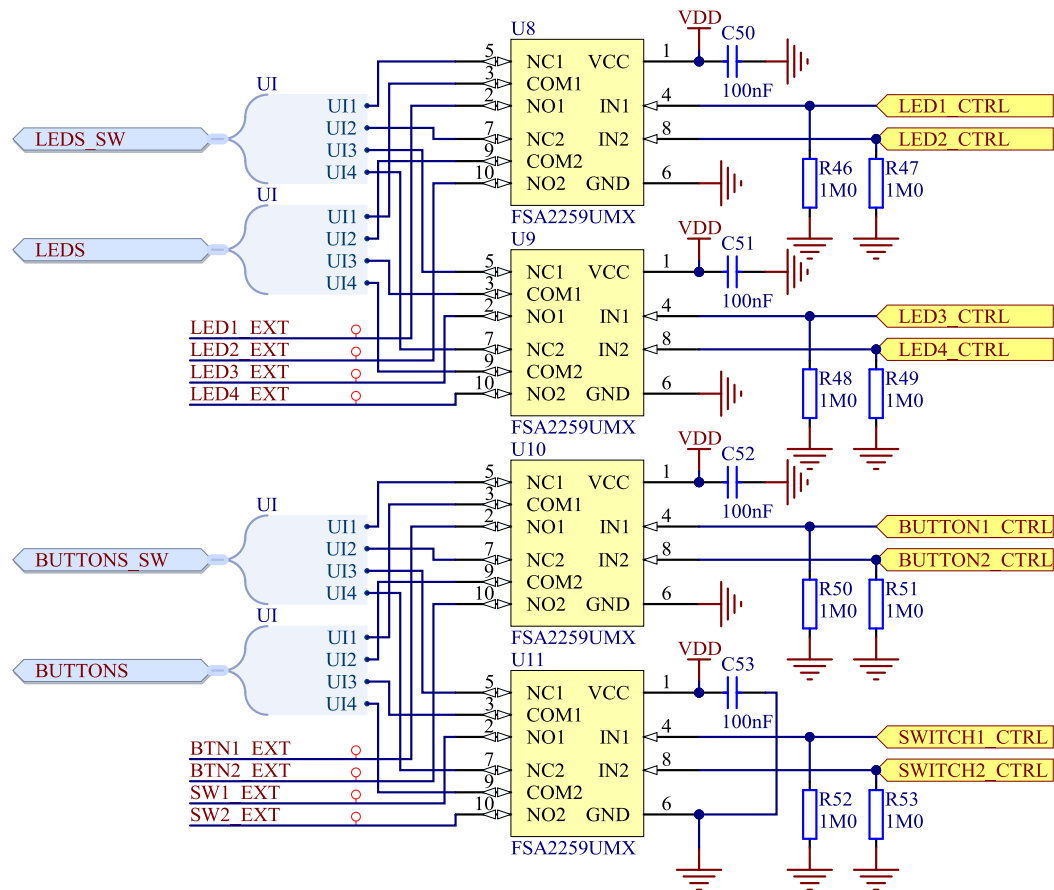


Figure 30: LED switch control

4.12.4 Switches for nRF52840 interface

Four analog switches control the routing of signals between the nRF9160 SiP and the nRF52840 SoC. There are ten signals that can be routed between the devices, seven *GPIOs* and three *COEX* pins.

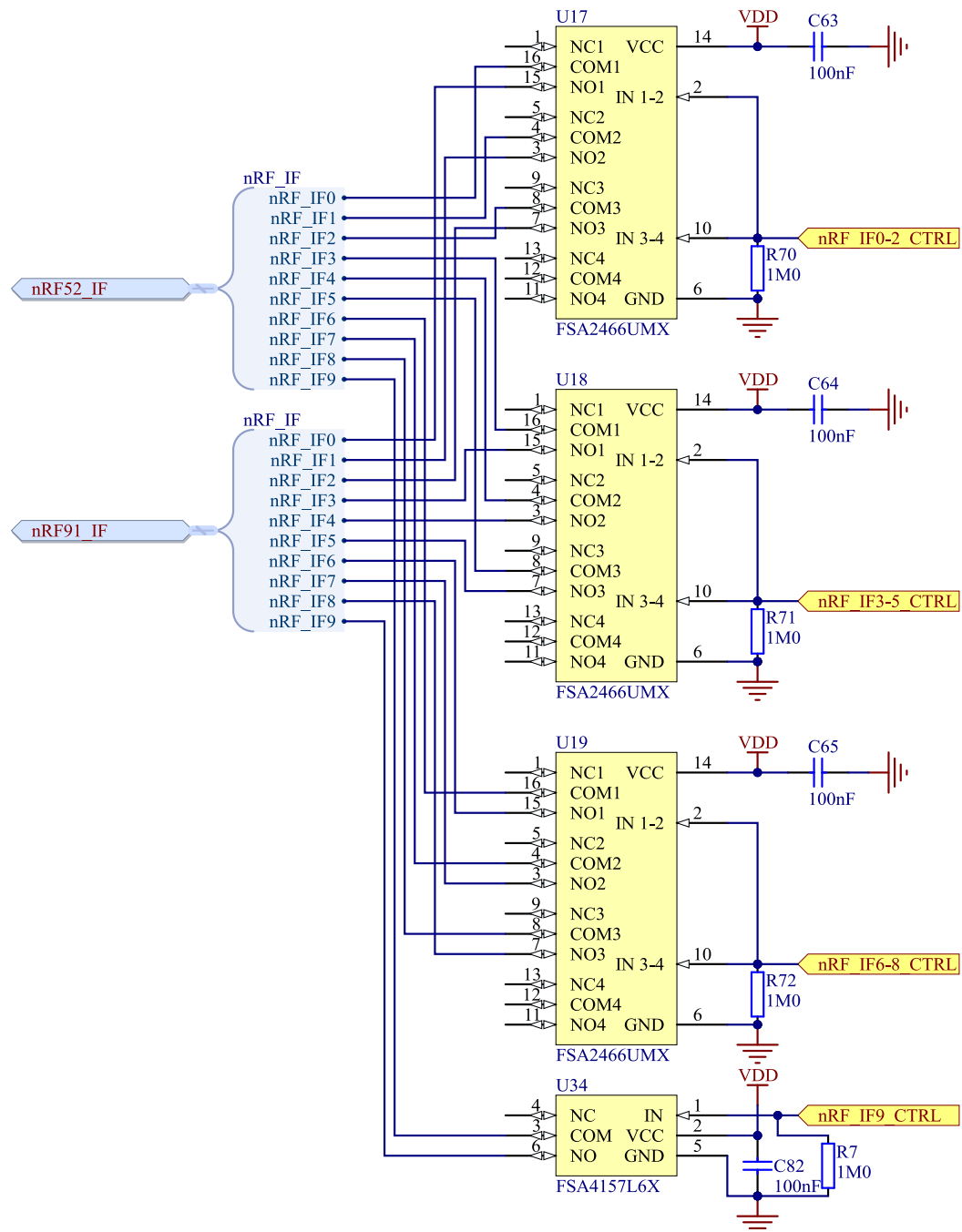


Figure 31: nRF52840 interface switches

4.12.5 Switches for external memory

Two analog switches are used to control the signals between the nRF9160 DK and the external memory.

The switches are controlled by **P0.19** of nRF52840.

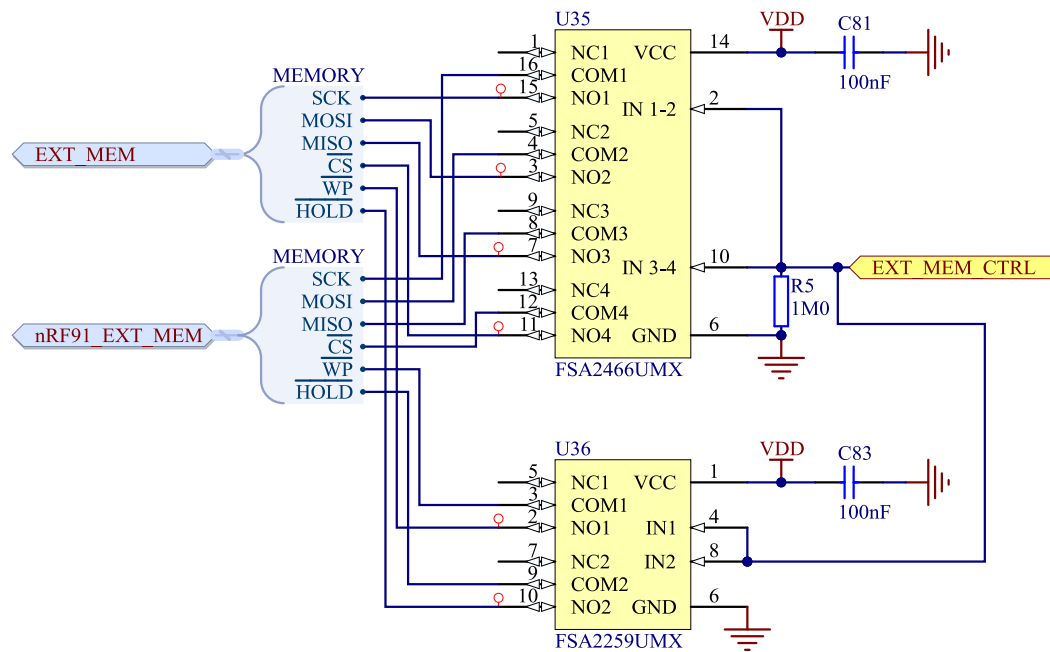


Figure 32: External memory switches

4.12.6 Switches for I/O expander

An analog switch is used to control the signals between the nRF9160 DK and the I/O expander.

The switch is controlled by **P0 . 14** of nRF52840.

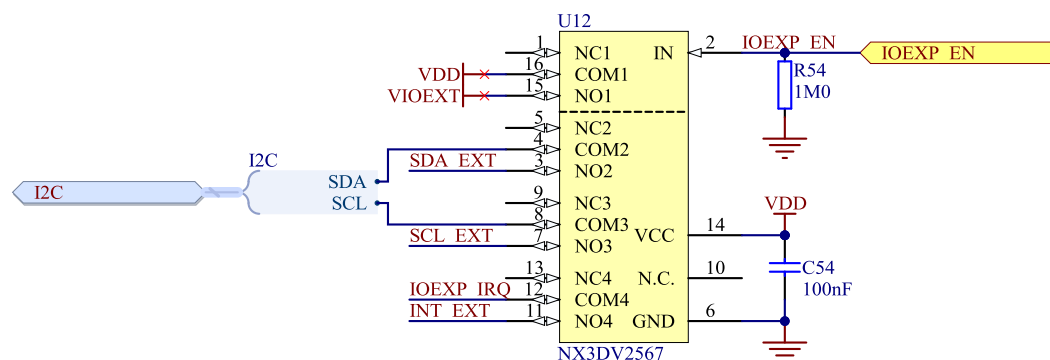


Figure 33: Switches for I/O expander

4.13 SIM and eSIM

The nRF9160 DK is designed to support regular *SIM* and *Embedded SIM (eSIM)*. For this purpose, it has a pluggable SIM card socket (**J5**) that takes a nano-sized SIM (4FF) and a non-populated footprint for an eSIM (MFF2).

Using the SIM socket is the default. If an eSIM is soldered onto the *DK*, the interface MCU application switch **SW7** can be used to select the eSIM. Connector **P28** can be used to connect and monitor the traffic on the SIM interface.

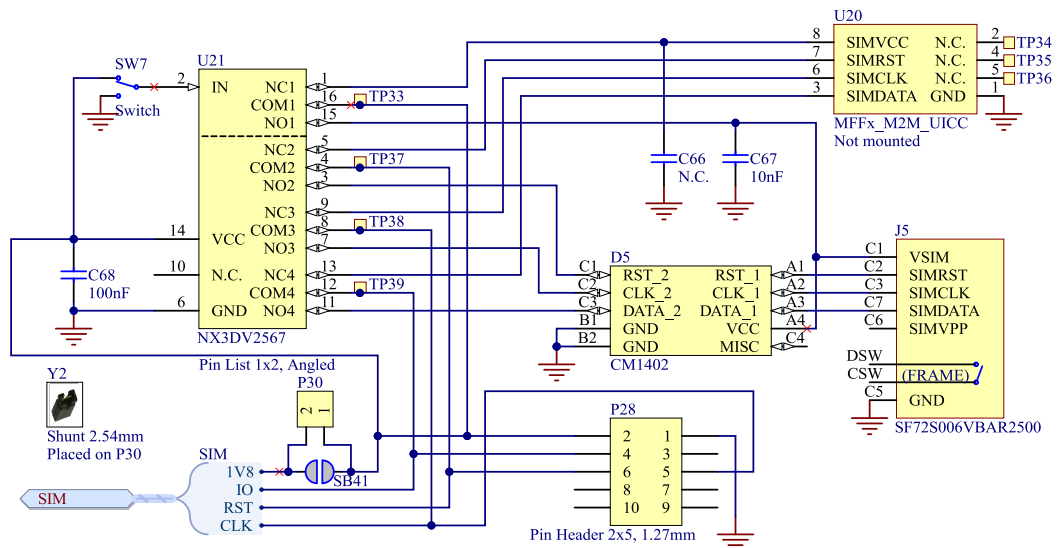


Figure 34: SIM card connector, eSIM, and selection switch

Note: The nano-SIM card is inserted with the electrical interface down in the SIM card holder (J5).

4.14 Additional interfaces

The nRF9160 DK supports dedicated interfaces for coexistence and RF control.

The nRF9160 DK has three COEX pins to control coexistence between multiple wireless devices. These pins are routed to nRF52840 for use when nRF52840 is used as a network processor. These signals are also available on connector **P24**. In addition, COEX0 is used to enable the GNSS LNA of the nRF9160 DK.

The nRF9160 DK has an interface allowing the modem to control external RF applications, such as antenna tuner devices. There are three MIPI RFFE interface pins, namely VIO, SCLK, SDATA, and three MAGPIO interface pins, namely MAGPIO0, MAGPIO1, and MAGPIO2. The MAGPIO and MIPI RFFE pins are not in use in this DK. The signals are accessed through test points listed in the following table.

Signal	Testpoint
SDATA	TP2
SCLK	TP3
VIO	TP4
MAGPIO0	TP5
MAGPIO1	TP6
MAGPIO2	TP7

Table 12: RF control interface testpoints

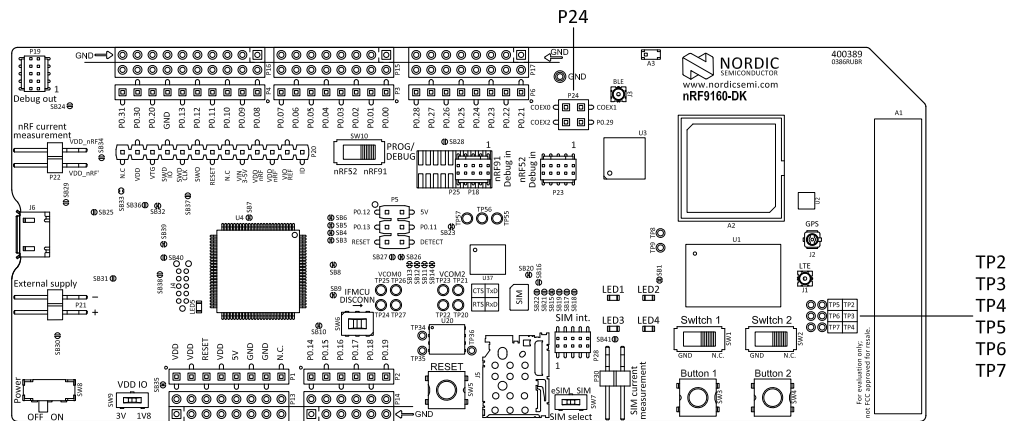


Figure 35: SiP-external interfaces

4.15 SiP enable

The nRF9160 SiP can be enabled by pulling pin **101** high or disabled by pulling pin **101** low. By default, the enable signal is pulled high by resistor **R1**.

4.16 Solder bridge configuration

The nRF9160 DK has a range of solder bridges for enabling or disabling functionality on the DK. Changes to these are not needed for normal use of the DK.

The following table is a complete overview of the solder bridges on the nRF9160 DK.

Solder bridge	Default	Function
SB1	Closed	Cut to disconnect VDD_GPIO supply to the nRF9160 SiP
SB2	Open	Short to connect DBG_CMD to the SD card reader J8 (not populated by default)
SB3	Closed	Cut to disconnect RTS signal from the nRF52840 SoC
SB4	Closed	Cut to disconnect CTS signal from the nRF52840 SoC
SB5	Closed	Cut to disconnect RXD signal from the nRF52840 SoC
SB6	Closed	Cut to disconnect TXD signal from the nRF52840 SoC
SB7	Closed	Cut to disconnect the 3.0 V USB supply from the interface MCU (U4)
SB8	Closed	Cut to disconnect SWDIO from the nRF52840 SoC
SB9	Closed	Cut to disconnect SWDCLK from the nRF52840 SoC
SB10	Closed	Cut to disconnect the RESET signal from the nRF52840 SoC
SB11	Closed	Cut to disconnect CTS from the nRF9160 SiP
SB12	Closed	Cut to disconnect RTS from the nRF9160 SiP
SB13	Closed	Cut to disconnect RXD from the nRF9160 SiP
SB14	Closed	Cut to disconnect TXD from the nRF9160 SiP

Solder bridge	Default	Function
SB15	Closed	Cut to disconnect the RESET button from the nRF9160 RESET pin when the interface MCU is disconnected
SB16	Open	Short to connect the RESET button to the RESET pin on the Arduino interface
SB17	Open	Short to connect the RESET pin on the Arduino interface to the nRF9160 reset pin
SB18	Open	Short to connect the RESET pin on the Arduino interface to the nRF9160 reset pin when the interface MCU is disconnected
SB19	Open	Short to connect the RESET pin on the Arduino interface to the interface MCU Boot when the interface MCU is disconnected
SB20	Closed	Cut to disconnect IOEXP_IRQ from nRF9160 GPIO when the I/O expander is enabled
SB21	Open	Short to connect the RESET button to the nRF52840 RESET pin when the interface MCU is disconnected
SB22	Open	Short to connect the RESET button to the RESET pin of the nRF52840
SB23	Closed	Cut to disconnect external memory chip select from P0.25
SB24	Open	Short to enable power supply of the external device when using the Debug out connector (P19)
SB25	Open	Short to bypass the USB detect switch (Q7)
SB26	Closed	Cut to permanently disable SHIELD DETECT feature
SB27	Open	Short to permanently enable the SHIELD DETECT feature
SB28	Open	Short to supply VDD to bottom UART debug connector
SB29	Open	Short to bypass the USB supply switch (U24)
SB30	Open	Short to bypass the external supply switch U25
SB31	Open	Short to bypass the VIN3–5 V switch U26
SB32	Closed	Cut to disconnect supply to LDO (U28)
SB33	Closed	Cut to disconnect the VSUPPLY and VDD_nRF'
SB34	Open	Short to connect VDD_nRF and VDD_nRF'
SB35	Closed	Cut to disconnect VDD and VIO
SB36	Open	Short to bypass the VDD_IMCU switch (U30)
SB37	Open	Short to bypass the VDD_IMCU_3V3 switch (U31)
SB38	Open	Short to bypass switch U33
SB39	Closed	Cut to disconnect supply from Arduino interface
SB40	Open	Short to bypass Arduino 5 V reverse protection diode D11
SB41	Open	Short to permanently connect SIM VDD, bypass SIM current measurement connector

Table 13: Solder bridge configuration

Current measurement

The current drawn by the nRF9160 *SiP* can be monitored on the nRF9160 *DK*.

Current can be measured using any of the following test instruments:

- Power analyzer
- Oscilloscope
- Ampere meter
- Power Profiler Kit II

Power analyzer measurements are not described in this document. For more information on the other instruments, see the following sections and [Power Profiler Kit II User Guide](#).

You can use connector **P22** for measuring current consumption or monitoring voltage levels on the nRF9160 DK. For more information, see [Prepare the DK for current measurements](#) on page 41.

The use of a USB connector is not recommended for powering the DK during current measurements due to potential noise from the USB power supply. Instead, the DK should be powered by connecting a power supply to connector **P21**.

For more information on measuring, see [Measure current profile with an oscilloscope](#) on page 42 and [Measure average current with an ampere meter](#) on page 43.

5.1 Prepare the DK for current measurements

To measure the current consumption of the *SiP*, you must first remove the jumper from **P22**.

Removing the jumper disconnects the nRF9160 SiP from the DK's power supply.

To restore normal kit functionality after measurement, apply the jumper on **P22** or short **SB34**.

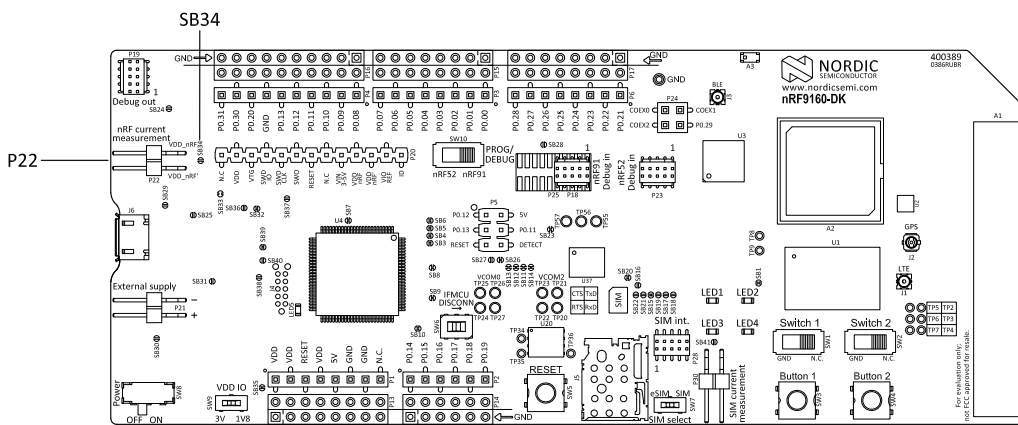


Figure 36: Solder bridge SB34 and P22 on the nRF9160 DK

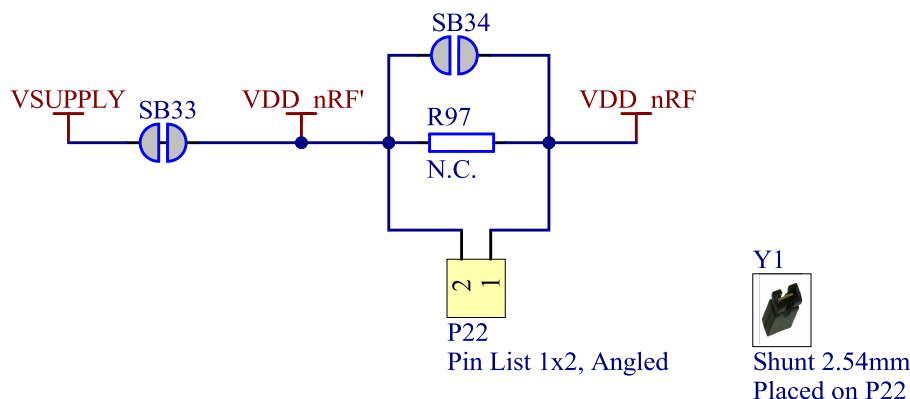


Figure 37: nRF power source select

5.2 Measure current profile with an oscilloscope

An oscilloscope can be used to measure the current over a given or continuous time interval and to capture the current profile.

Before you start, make sure you have prepared the DK as described in [Prepare the DK for current measurements](#) on page 41.

Complete the following steps to measure the current profile of the nRF9160 DK:

1. Solder a $0.5\ \Omega$ resistor to **R97**.
2. Set the oscilloscope to differential mode or to a mode that is similar.
3. Connect the oscilloscope using two probes on the pins of the **P22** connector.
4. Calculate or plot the instantaneous current from the voltage drop across the $0.5\ \Omega$ resistor by taking the difference of the voltages measured on the two probes.

The voltage drop is proportional to the current.

5. Measure the voltage drop over the $0.5\ \Omega$ resistor to get the current power profile.

The plotted voltage drop can be used to calculate the current at a given point in time. The current can then be averaged or integrated to analyze current and energy consumption over a period.

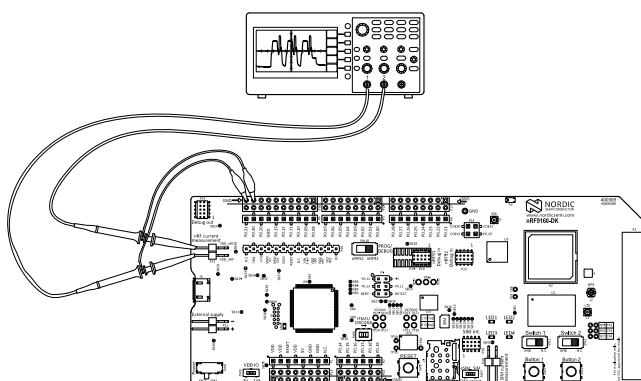


Figure 38: Current measurement with an oscilloscope

To reduce noise, do the following:

- Use probes with 1x attenuation.
- Enable averaging mode to reduce random noise.
- Enable high resolution function if available.

A minimum of one sample every $5\ \mu\text{s}$ is needed to accurately measure the average current.

5.3 Measure average current with an ampere meter

The current drawn by the nRF9160 SiP can be measured using an ampere meter.

Before you start, make sure you have prepared the DK as described in [Prepare the DK for current measurements](#) on page 41.

Complete the following steps to measure the average current drawn by the nRF9160 SiP:

1. Set the average timing of the ampere meter to a long interval, such as 1 s or longer.
2. Set the dynamic range of the ampere meter between 1 μ A and 15 mA so that it is wide enough to provide accurate measurements.
3. To connect the ampere meter in series with the nRF9160 DK, connect the ampere meter to the pins on connector **P22**.

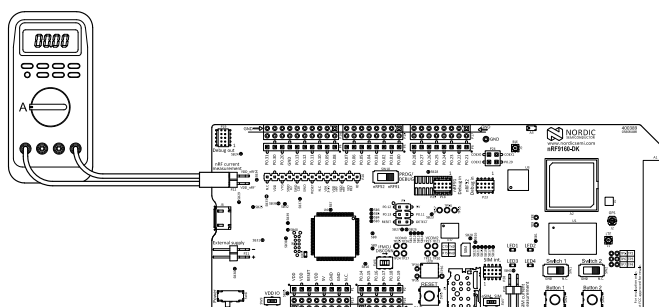


Figure 39: Current measurement with an ampere meter

Note: Use a high-speed, high-dynamic-range ampere meter for the best and most reliable measurements. Switching the current range in the ampere meter can affect the power supply to the nRF9160 DK. High speed and bandwidth are required to detect rapid changes in the current consumption in the nRF9160 DK.

6 RF measurements

The nRF9160 DK is equipped with a small coaxial connector **J1** for measuring the LTE RF signal and **J3** for 2.4 GHz.

The connectors are of *Microwave coaxial connector with switch (SWF)* type (Murata part no. MM8130-2600) with an internal switch. By default, when no cable is attached, the RF signal is routed to the onboard antenna. The insertion loss in the adapter cable is approximately 0.5 dB–1 dB.

An adapter (Murata part no. MXHS83QE3000) is available with a standard SMA connection on the other end for connecting instruments (the adapter is not included in the kit). When connecting the adapter, the internal switch in the SWF connector disconnects the onboard antenna and connects the RF signal from nRF9160 or nRF52840 to the adapter.

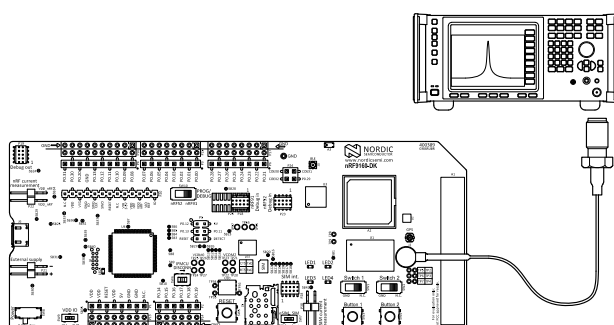


Figure 40: Connecting a spectrum analyzer

7 Radiated performance

The *LTE* antenna on the *DK* is optimized for global operation, supporting all *LTE* frequency bands in the region of 698 MHz to 960 MHz and 1710 MHz to 2200 MHz.

All antennas on the nRF9160 DK have fixed matching networks, which means that a matching configuration is not needed to switch between the frequency bands. The average performance of the *LTE* antenna mounted on the nRF9160 DK is shown in the following table. The *LTE* antenna also supports other frequency ranges, but it is not optimized for operating on these frequency bands.

Band	Average return loss (dB)	Average efficiency (%)
3	-13.5	62.0
4	-7.1	47.1
13	-8.5	58.5
20	-10.5	73.8

Table 14: Antenna performance for corresponding radio band

The 2.4 GHz antenna on the nRF9160 DK is optimized for the frequency range of 2.4 GHz to 2.48 GHz. It has an average return loss of -15 dB, average efficiency being 80%.

Glossary

AT command

A command used to control the modem.

Band-Pass Filter (BPF)

An electronic device or circuit that passes frequencies within a certain range and rejects frequencies outside that range.

Cat-M1

LTE-M User Equipment (UE) category with a single RX antenna, specified in 3GPP Release 13.

Cat-NB1

NB-IoT User Equipment (UE) category with 200 kHz UE bandwidth and a single RX antenna, specified in 3GPP Release 13.

Cat-NB2

An upgraded version of Cat-NB1, specified in 3GPP Release 14.

Clear to Send (CTS)

In flow control, the receiving end is ready and telling the far end to start sending.

DECT NR+

A non-cellular radio standard included as part of the 5G standards by the ITU.

DC

Direct Current

Development Kit (DK)

A hardware development platform used for application development.

Electrostatic Discharge (ESD)

A sudden discharge of electric current between two electrically charged objects.

Embedded SIM (eSIM)

A form of programmable *SIM* that is embedded directly into a device.

Fast Identity Online (FIDO)

A bundle of hardware and software modules serving as a tracing interface between traced device(s) and user application.

Global Navigation Satellite System (GNSS)

A satellite navigation system with global coverage. The system provides signals from space transmitting positioning and timing data to GNSS receivers, which use this data to determine location.

General-Purpose Input/Output (GPIO)

A digital signal pin that can be used as input, output, or both. It is uncommitted and can be controlled by the user at runtime.

Global Positioning System (GPS)

A satellite-based radio navigation system that provides its users with accurate location and time information over the globe.

Hardware Flow Control (HWFC)

A handshaking mechanism used to prevent an overflow of bytes in modems. It uses two dedicated pins on the RS-232 connector, Request to Send and Clear to Send.

Integrated Circuit (IC)

A semiconductor chip consisting of fabricated transistors, resistors, and capacitors.

Inter-integrated Circuit (I²C)

A multi-master, multi-slave, packet-switched, single-ended, serial computer bus.

Low-Dropout Regulator (LDO)

A linear voltage regulator that can operate even when the supply voltage is very close to the desired output voltage.

Low-Noise Amplifier (LNA)

In a radio receiving system, an electronic amplifier that amplifies a very low-power signal without significantly degrading its signal-to-noise ratio.

Low-Power Wide Area (LPWA)

A wireless communication technology designed to allow long-range communication at a low bit rate.

Long-Term Evolution (LTE)

A wireless broadband communication standard for mobile devices and data terminals, based on the GSM/EDGE and UMTS/HSPA technologies.

LTE-M

An open standard that is most suitable for medium throughput applications requiring low power, low latency, and/or mobility, like asset tracking, wearables, medical, POS, and home security applications. Also known as Cat-M1.

Mass Storage Device (MSD)

Any storage device that makes it possible to store and port large amounts of data in a permanent and machine-readable fashion.

nRF Cloud

Nordic Semiconductor's platform for connecting IoT devices to the cloud, viewing and analyzing device message data, prototyping ideas that use Nordic Semiconductor chips, and more. It includes a public REST API that can be used for building IoT solutions. See [nRF Cloud portal \(nrfcloud.com\)](https://nrfcloud.com).

Operational Amplifier (op-amp)

A high-gain voltage amplifier that has a differential input and, usually, a single output.

Printed Circuit Board (PCB)

A board that connects electronic components.

Receive Data (RXD)

A signal line in a serial interface that receives data from another device.

Request to Send (RTS)

In flow control, the transmitting end is ready and requesting the far end for a permission to transfer data.

SAW filter

A high-performing filter using Surface Acoustic Wave (SAW) technology. This technology employs piezoelectric transducers, which, when excited, produce waves that are used to filter out desired frequencies.

Serial Wire Debug (SWD)

A standard two-wire interface for programming and debugging Arm® CPUs.

Subscriber Identity Module (SIM)

A card used in User Equipment (UE) containing data for subscriber identification.

Surface Acoustic Wave (SAW)

An acoustic wave traveling along the surface of a material exhibiting elasticity, with an amplitude that typically decays exponentially with depth into the substrate.

Subscriber Identity Module (SIM)

A card used in User Equipment (UE) containing data for subscriber identification.

System in Package (SiP)

Several integrated circuits, often from different technologies, enclosed in a single module that performs as a system or subsystem.

System on Chip (SoC)

A microchip that integrates all the necessary electronic circuits and components of a computer or other electronic systems on a single integrated circuit.

Microwave coaxial connector with switch (SWF)

A small, RF surface-mount switch connector series for wireless applications.

Serial Wire Output (SWO)

A data line for tracing and logging.

Transmit Data (TXD)

A signal line in a serial interface that transmits data to another device.

Universal Asynchronous Receiver/Transmitter (UART)

A hardware device for asynchronous serial communication between devices.

Universal Integrated Circuit Card (UICC)

A new generation Subscriber Identity Module (SIM) used in User Equipment (UE) for ensuring the integrity and security of personal data.

User Equipment (UE)

Any device used by an end-user to communicate. The UE consists of the Mobile Equipment (ME) and the Universal Integrated Circuit Card (UICC).

U.FL

An ultra-small surface-mount coaxial connector designed for high-frequency performance.

Universal Serial Bus (USB)

An industry standard that establishes specifications for cables and connectors and protocols for connection, communication, and power supply between computers, peripheral devices, and other computers.

Recommended reading

In addition to the information in this document, you might need to consult other documents.

Nordic documentation

- [Getting started with nRF9160 DK](#)
- [nRF9160 Product Specification](#)
- [nRF9160 Modem Firmware Release Notes](#) (included in the latest [nRF9160 modem firmware](#))
- [nRF Connect SDK documentation](#)
- [nRF9160 AT Commands](#)
- [nRF Connect Programmer](#)

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